Signal and System Design (IGS2137)

Jump Together, Fly Farther!

Weeks 3&4 Lectures

School of Global Convergence Studies

인하대학교 국제학부

ISE Department

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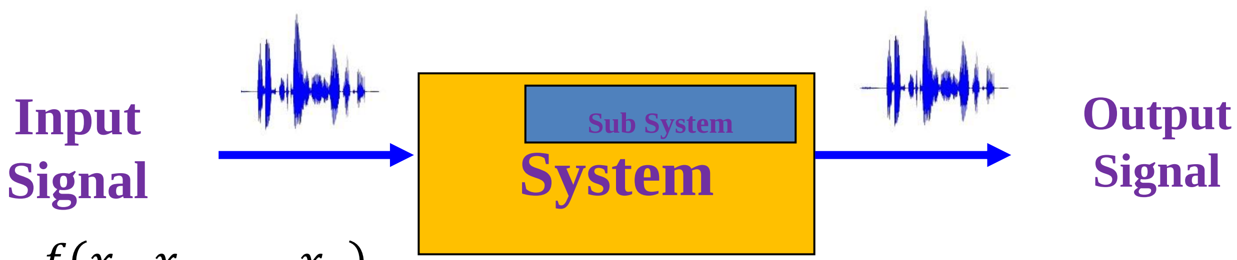
Course Plan

- Introduction (Ch1)
- Basic Definitions of Signal and systems (Ch1)
- Time Domain of LTIV Systems (Ch2)
- Frequency Domain of Signal & LTIV Systems (Ch3)
- Applications to Mixed Signals (Ch4)
- Laplace Transform: Continuous Complex Exponentials (Ch6)
- Z-Transform: Discrete Complex Exponentials (CH11)

What is a Signal and a System?

- Signal: A signal is formally defined as a *function* of one or more variables, which contains information on the nature of the physical phenomenon.

  - functions of *Time, Frequency, Space,* etc.

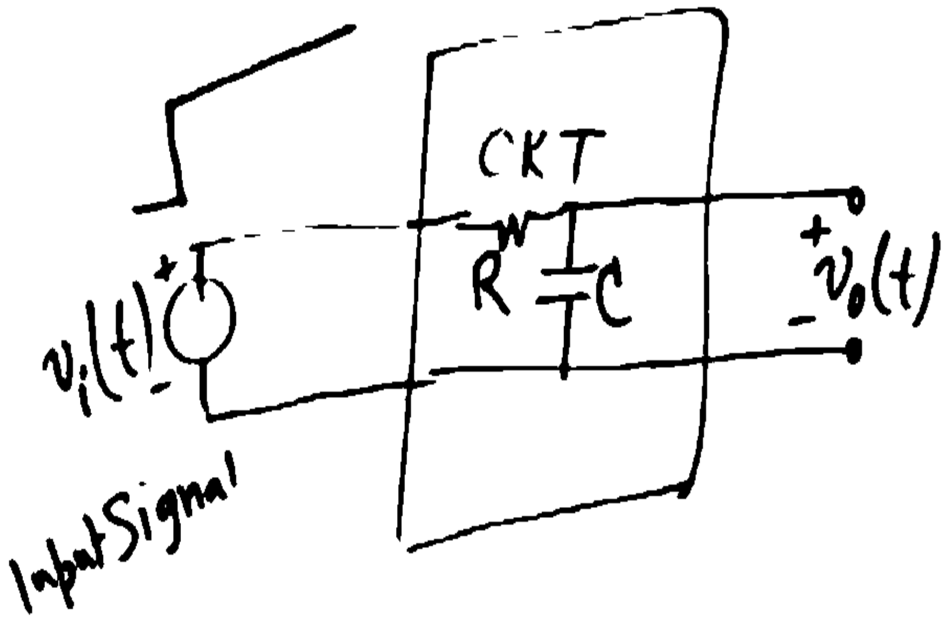


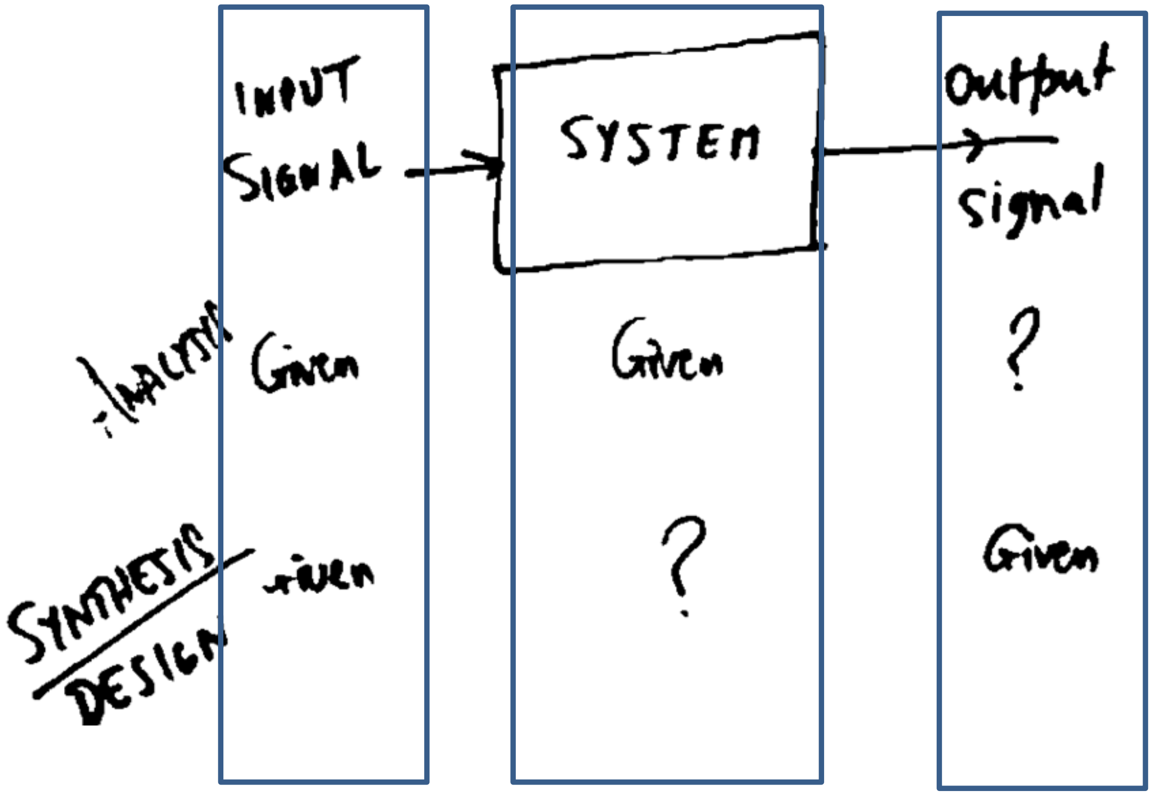
$f(x_1, x_2, \ldots, , x_n)$

$x(t)$

- System : A system is formally defined as an entity that manipulates one or more signals to accomplish a function, thereby yielding new signals.

Some Interesting Systems




INPUT SIGNAL
SYSTEM
Output signal
Given
Given
?
SYNTHESIS
DESIGN
?
Given

ANALYSIS TOOLS

CIRCUIT THEORY

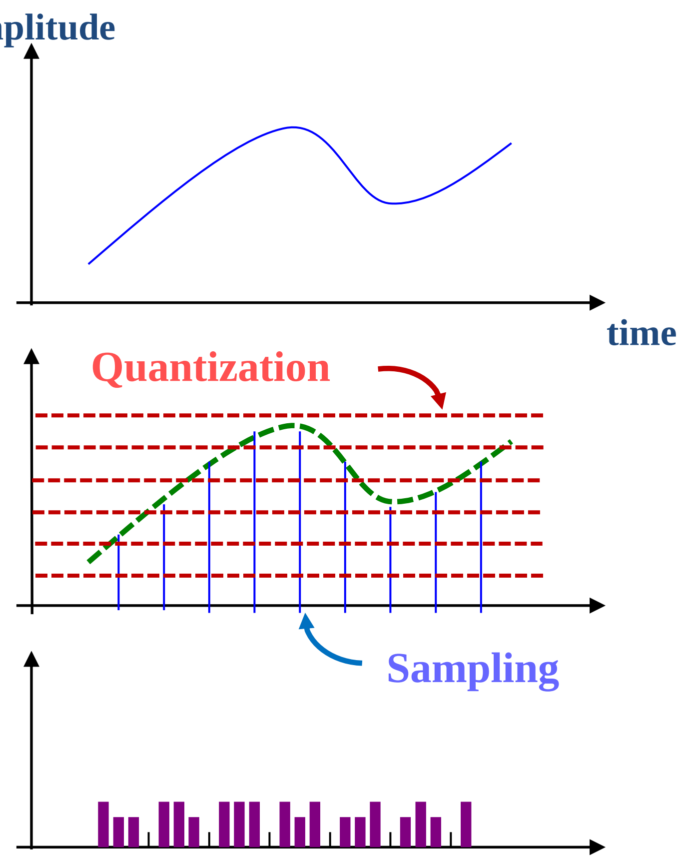
CONTROL ENGG

COMMUNICATION SYSTEMS

Digital Signals

ADC
(Analog to digital convertor)

Analog Signal $x(t)$ → **Sampling** → Discrete-time Signal $x[n]$ → **Quantization** → Digital Signal



Analog signal
(Continuous-time Signal)

Sampling and quantization

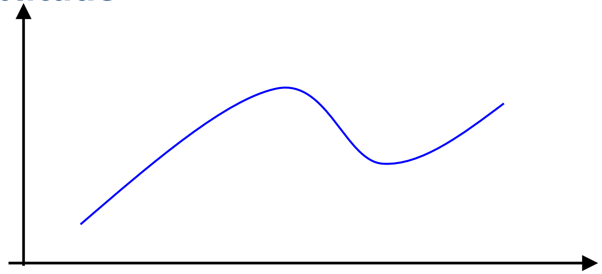
Discrete-time signal

coding

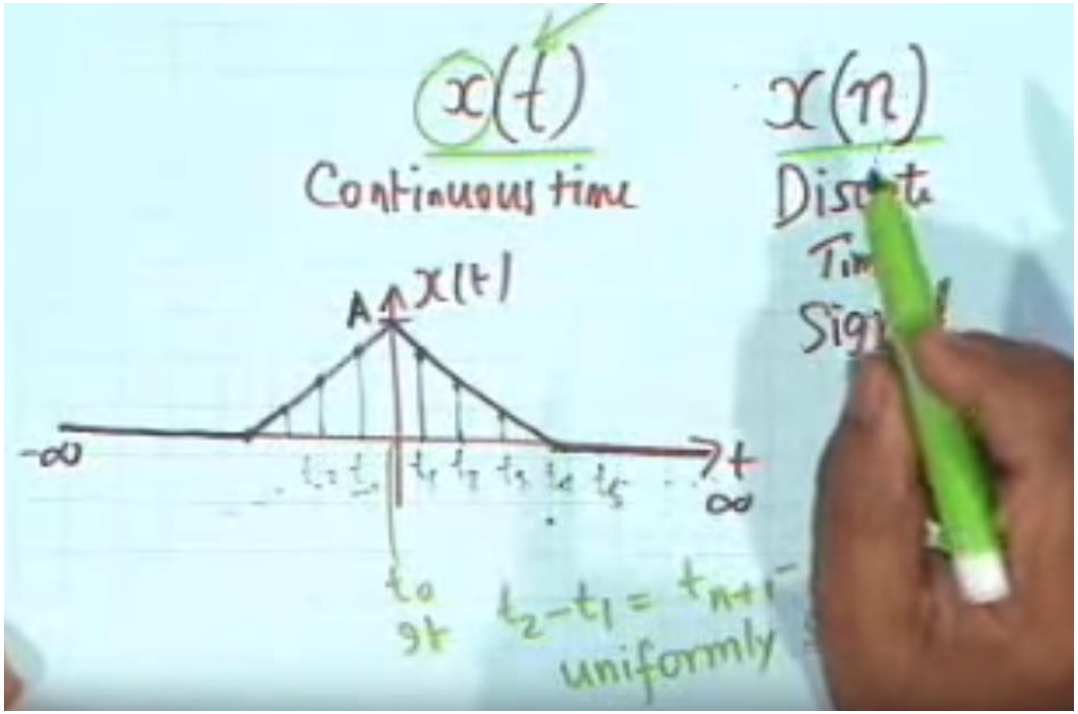
Digital Signal

Digital Signals

amplitude

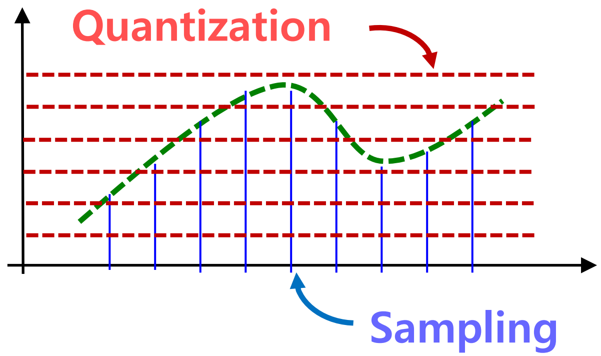


Analog signal
(Continuous-time Signal)

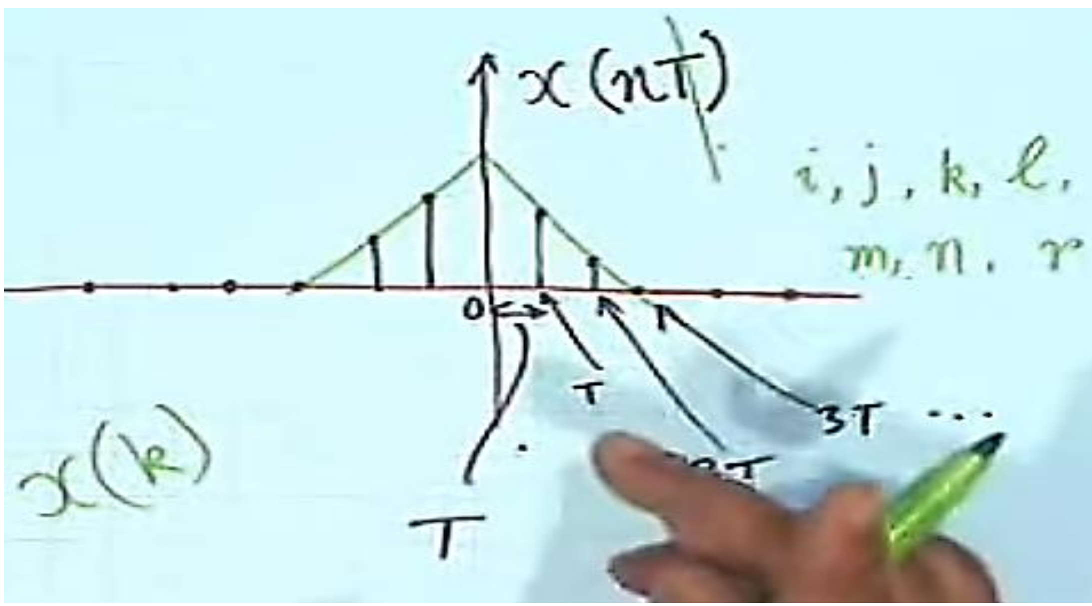


Digital Signals

Sampling and quantization



Discrete-time signal

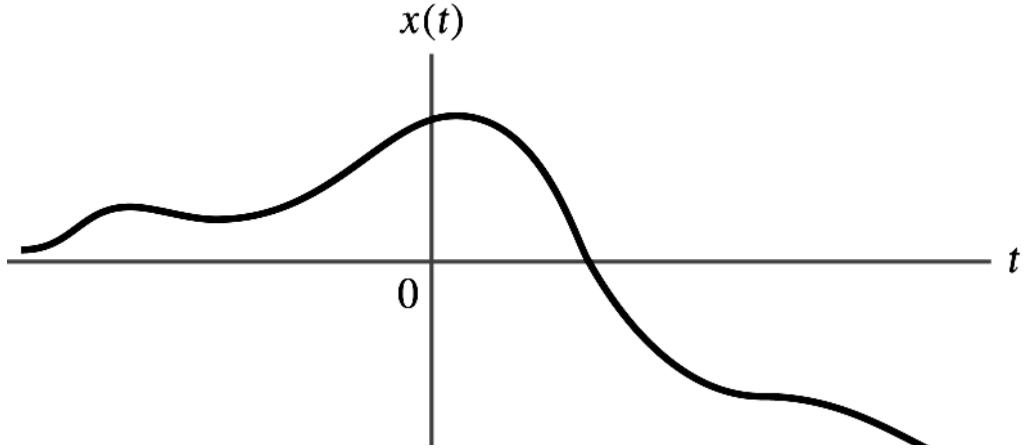


where $t = nT_s$

$$x[n] = x(nT_s), \quad n = 0, \ \pm 1, \ \pm 2, \ \ldots\ldots$$

Digital Signals

Figure 1.11 (p. 17)
Continuous-time signal.



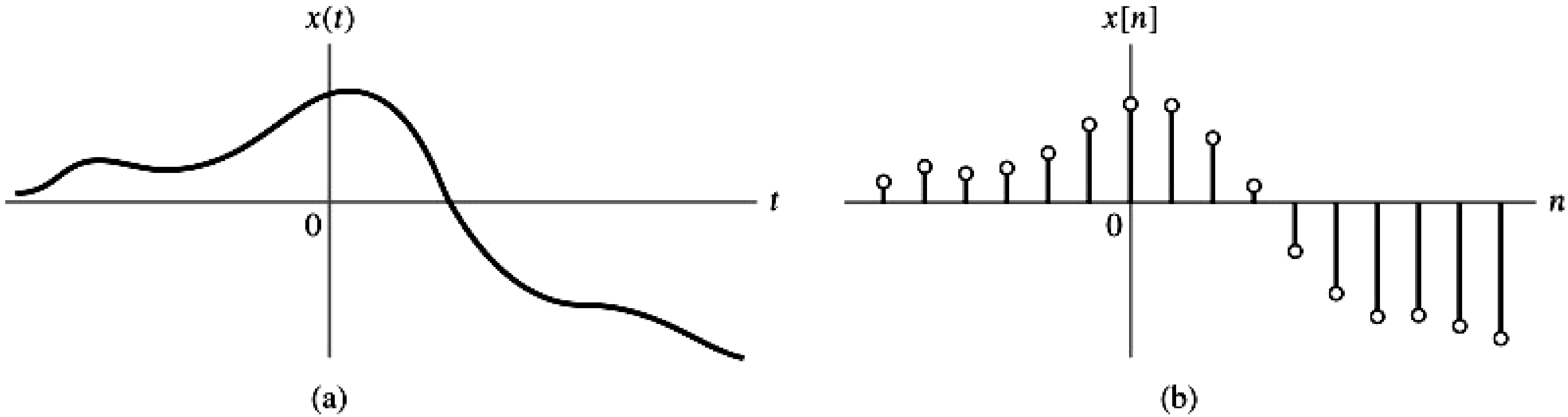


Figure 1.12 (p. 17)
(a) Continuous-time signal *x*(*t*). (b) Representation of *x*(*t*) as a discrete-time signal *x*[*n*].

Digital Signals : Folding Reflection

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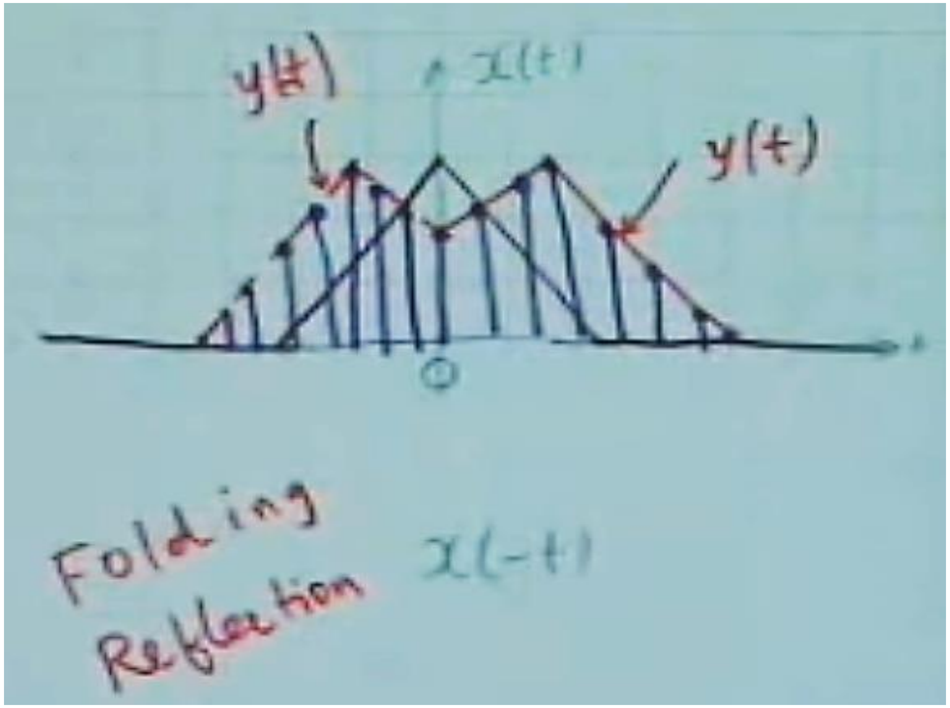
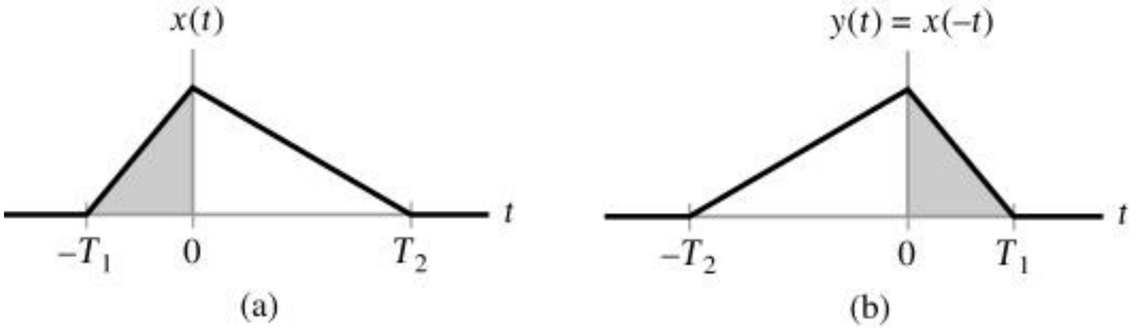


$$y(t) = x(-t)$$



Digital Signals : Time scaling

Fig. 1-20.

$y(t) = x(at)$ $\Rightarrow$ $\begin{cases} a > 1 \Rightarrow \text{compressed in time} \\ 0 < a < 1 \Rightarrow \text{expanded (stretching) in time} \end{cases}$

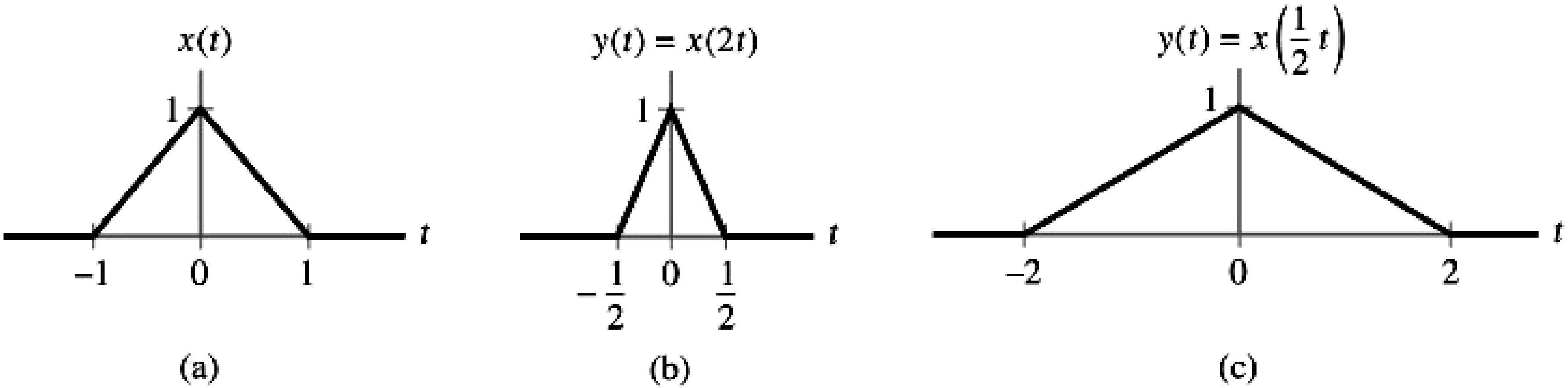


Figure 1.20 (p. 27)
Time-scaling operation; (a) continuous-time signal $x(t)$, (b) version of $x(t)$ compressed by a factor of 2, and (c) version of $x(t)$ expanded by a factor of 2.

Capacitor: $v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau$ (1.27)

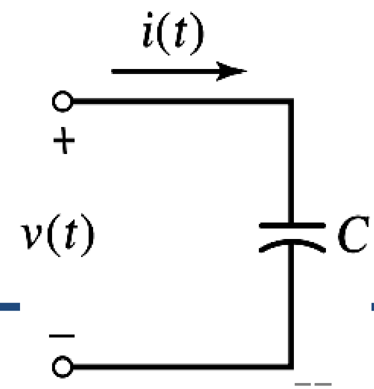
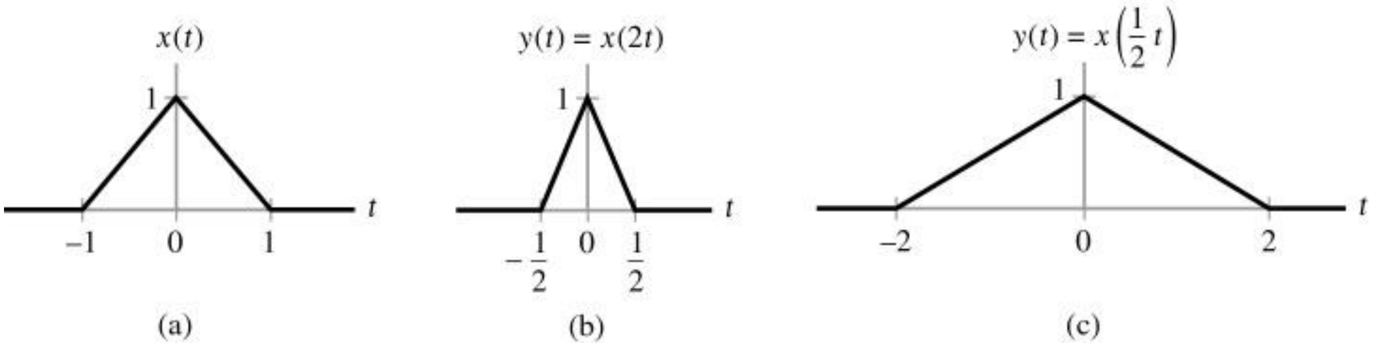


Figure 1.19 (p. 27)
Capacitor with voltage $v(t)$ across its terminals, inducing current $i(t)$.

Operations performed on independent signals

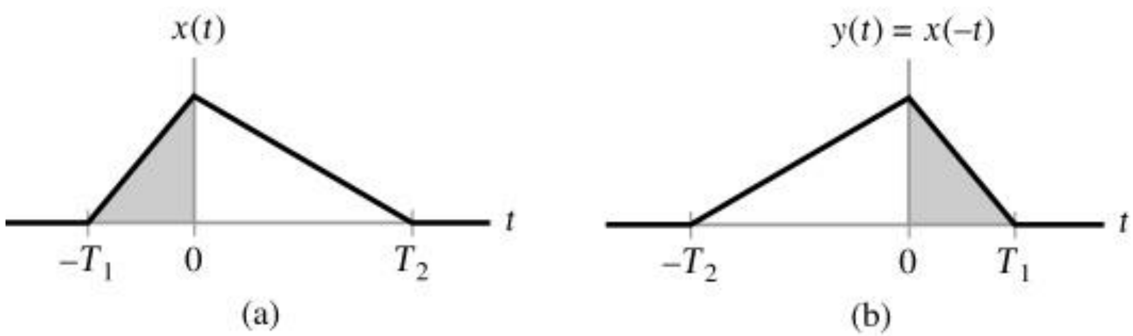
- **Time scaling**

$$y(t) = x(at), \quad \begin{cases} a > 1: & Compressed \\ 0 < a < 1: & Expanded \end{cases}$$



- **Reflection**

$$y(t) = x(-t)$$



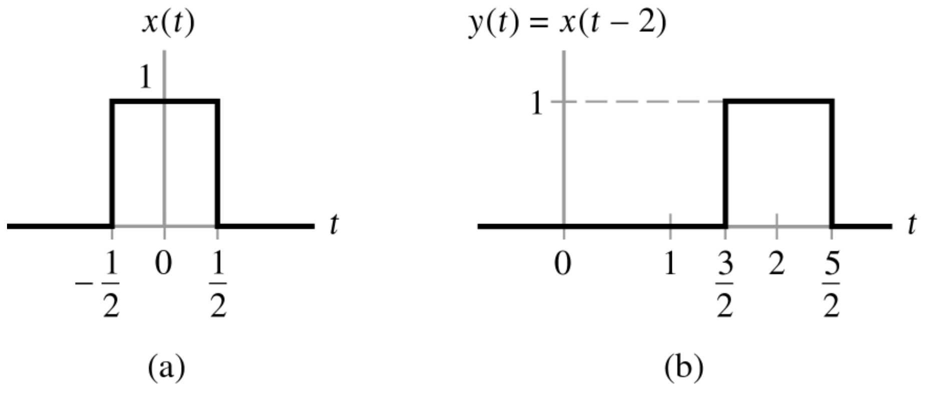
Operations performed on independent signals

- **Time shifting**

  - Precedence rule for time shifting & time scaling

$$y(t) = x(t - t_0)$$

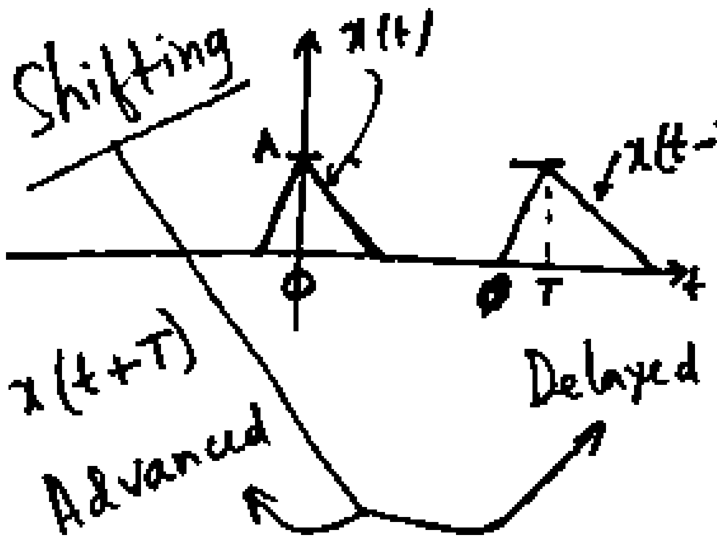
$$y(t) = x(at - b) = x\left(a(t - \frac{b}{a}) \right)$$



Operations performed on independent signals

- **Time shifting**

 - Precedence rule for time shifting



1.4. Signal Classification

- Deterministic signal:
- Random signal:

- Periodic signal
- Aperidic(Non-periodic) signal

- Power signal:
- Energy signal:

- Continuous signal:
- Discrete signal:

1.4. Signal Classification

The total energy of the continuous-time signal $x(t)$ is

$$E = \lim_{T\to\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt = \int_{-\infty}^{\infty} x^2(t)dt \qquad (1.15)$$

Time-averaged, or average, power is

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt \qquad (1.16)$$

For periodic signal, the time-averaged power is

$$P = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt \qquad (1.17)$$

◆ **Discrete-time case:**

Total energy of $x[n]$:

$$E = \sum_{n=-\infty}^{\infty} x^2[n] \qquad (1.18)$$

Average power of $x[n]$:

$$P = \lim_{n\to\infty} \frac{1}{2N} \sum_{n=-N}^{N} x^2[n] \qquad (1.19)$$

$$\Rightarrow \quad P = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

1.4. Signal Classification

Deterministic signals and random signals

A *deterministic signal* is a signal about which there is no uncertainty with respect to its value at any time.

Figure 1.13 ~ Figure 1.17

A *random signal* is a signal about which there is uncertainty before it occurs.

Figure 1.9

Energy signals and power signals

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} \quad (1.12)$$

$$p(t) = Ri^2(t) \quad (1.13)$$

If $R = 1\ \Omega$ and $x(t)$ represents a current or a voltage, then the instantaneous power is

$$p(t) = x^2(t) \quad (1.14)$$

1.4. Signal Classification

- Deterministic signal:
- Random signal:

- Periodic signal
- Aperidic(Non-periodic) signal

★ ***Power signal***: $0 < P < \infty$

If and only if the average power of the signal satisfies the condition

★ ***Energy signal***:

If and only if the total energy of the signal satisfies the condition

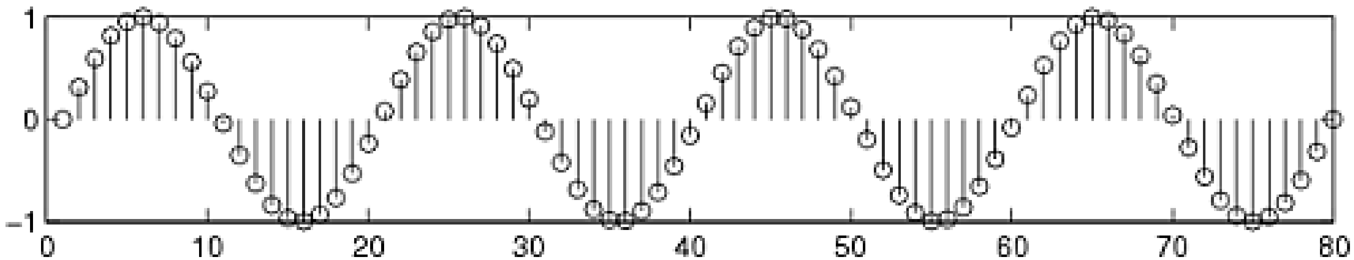
$0 < E < \infty$

- Continuous signal:
- Discrete signal:

1.4. Signal Classification

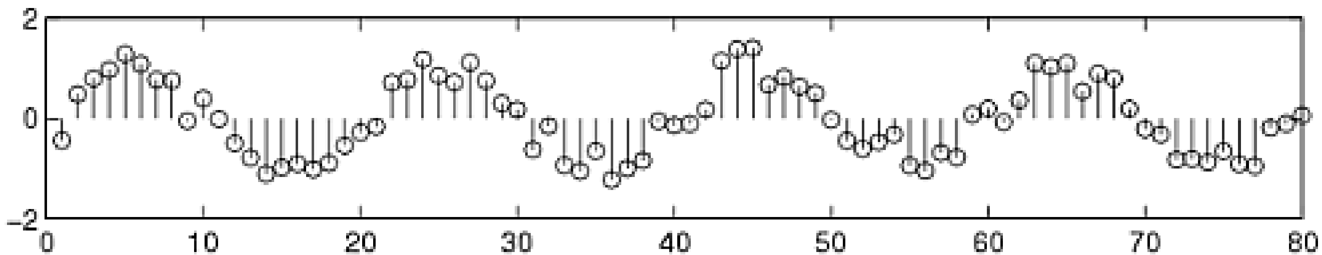
- Deterministic signal:

There is no uncertainty with respect to its value at any time. (ex) sin(3t)



- Random signal:

There is uncertainty before its actual occurrence.



1.4. Signal Classification

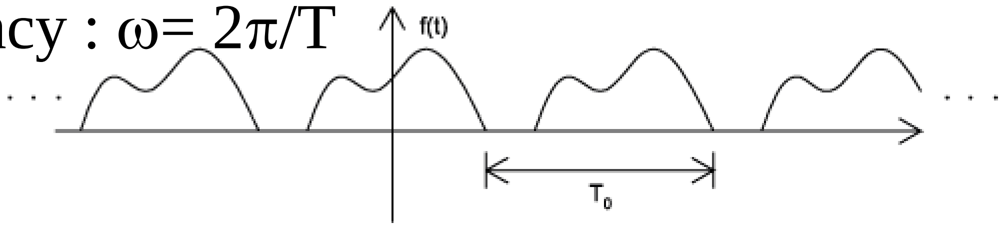
- Periodic signal

A function that satisfies the condition

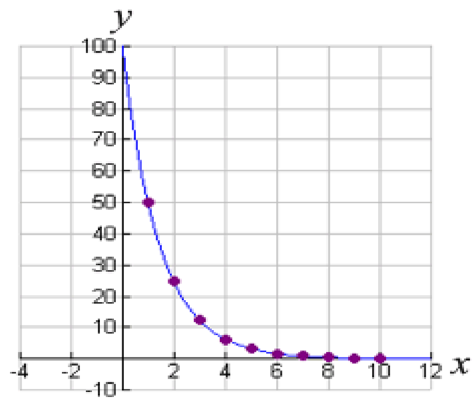
x(t)=x(t+T) for all t

Fundamental frequency : f=1/T

Angular frequency : ω= 2π/T



- Aperidic signal e



), etc

1.4. Signal Classification

- Energy signal: it has nonzero, but finite energy $0 < E_X < \infty$ for all t.

$$E_X = \lim_{T \to \infty} \int_{-T/2}^{T/2} x^2(t)dt = \int_{-\infty}^{\infty} x^2(t)dt$$

ex) aperiodic signal , exp(-at)

- Power signal: it has finite but nonzero power($0 < P_X < \infty$)

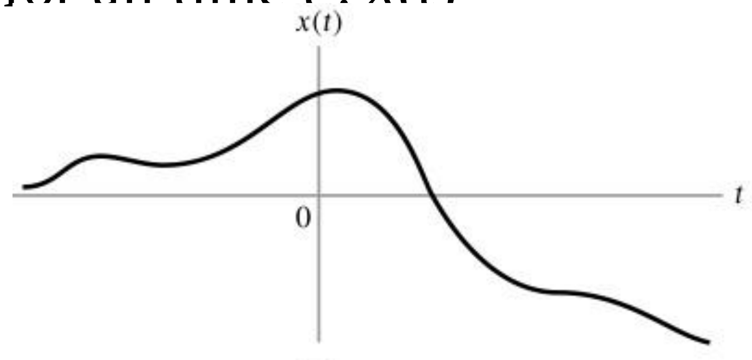
$$P_X = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)dt$$

ex) periodic signal

1.4. Signal Classification

- Continuous signal:

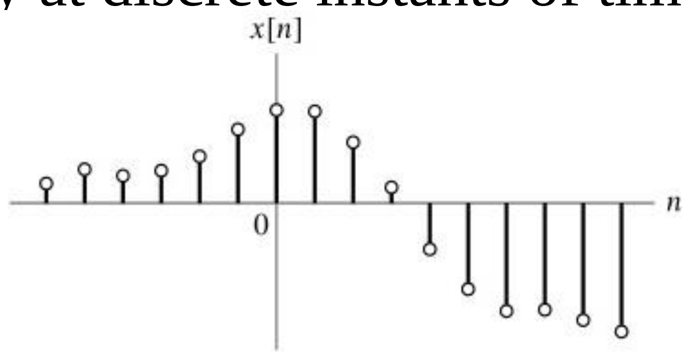
 - It is defined for all time t : $x(t)$



(a)

- Discrete signal: x(kT) , k: integer, T: fixed value

- It is defined only at discrete instants of time : $x[n]=x(nT)$

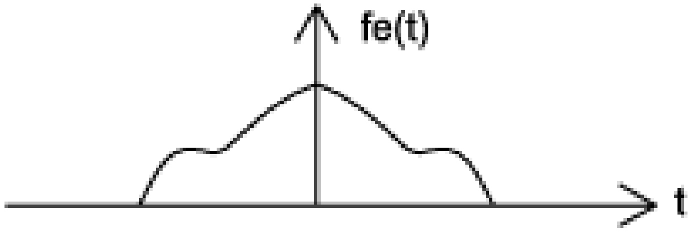


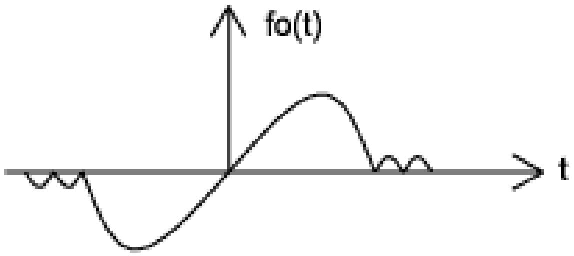
(b)

1.4. Signal Useful Backgrounds

- **Even and odd signals**
  - Even signals : $x(-t)=x(t)$
  - Odd signals : $x(-t)=-x(t)$
  - Even and odd signal decomposition

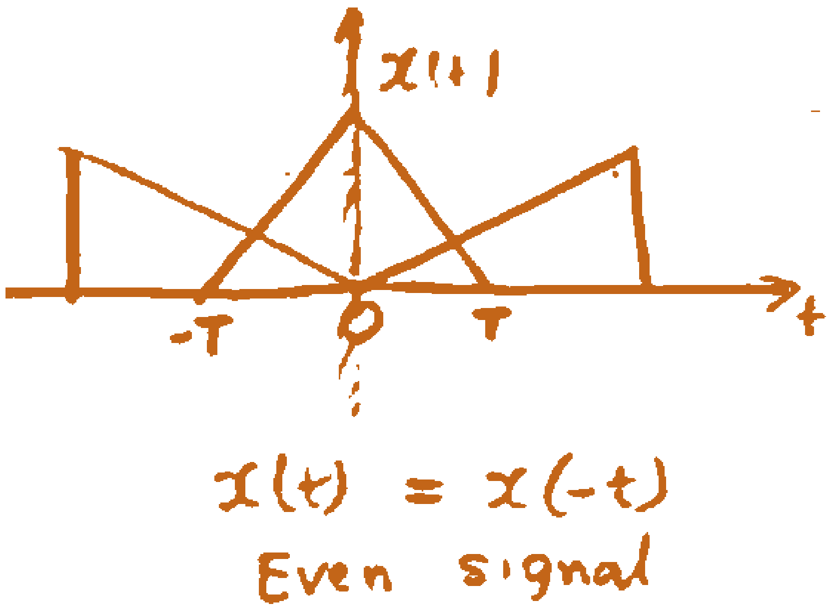
$x_e(t)= 1/2\cdot(x(t)+x(-t))$ $x_o(t)= 1/2\cdot(x(t)-x(-t))$

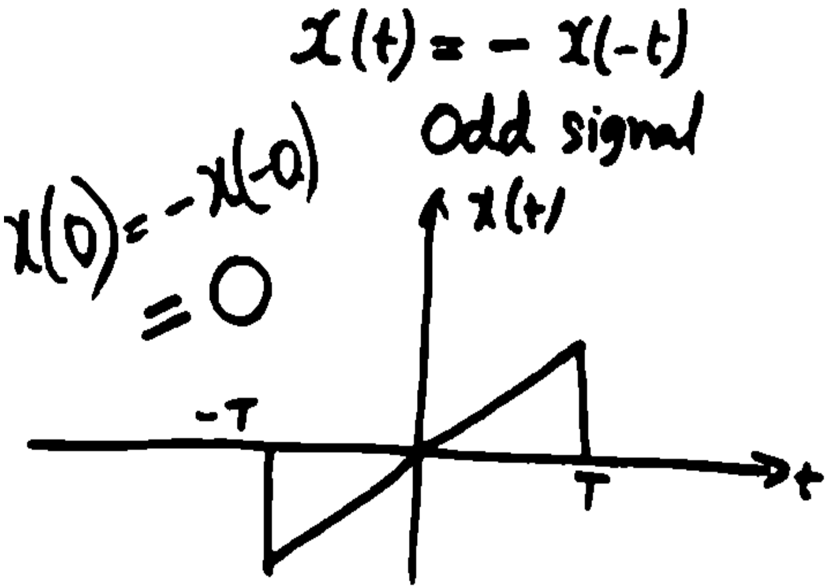




1.4. Signal Useful Backgrounds

- **Even and odd signals**





1.4. Signal Useful Backgrounds

- **Even and odd signals**

General $x(t)$

$$= Ev[x(t)] + Od[x(t)] \quad (1)$$

$$x(-t) = Ev[x(-t)] + Od[x(-t)]$$

$$= Ev[x(t)] - Od[x(t)] \quad (2)$$

$$\begin{matrix} Ev \\ Od \end{matrix}[x(t)] = \frac{x(t) \pm x(-t)}{2}$$

1.4. Signal Classification

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Periodic and nonperiodic signals (Continuous–Time Case)

Periodic signals: $x(t) = x(t+T) \quad \text{for} \quad \text{all} \quad t$ (1.7)

$T = T_0, \; 2T_0, \; 3T_0, \; \ldots\ldots$ and $T = T_0 \equiv \text{Fundamental} \quad \text{period}$

Fundamental frequency:

$$f = \frac{1}{T} \qquad (1.8)$$

Angular frequency:

$$\omega = 2\pi f = \frac{2\pi}{T} \qquad (1.9)$$

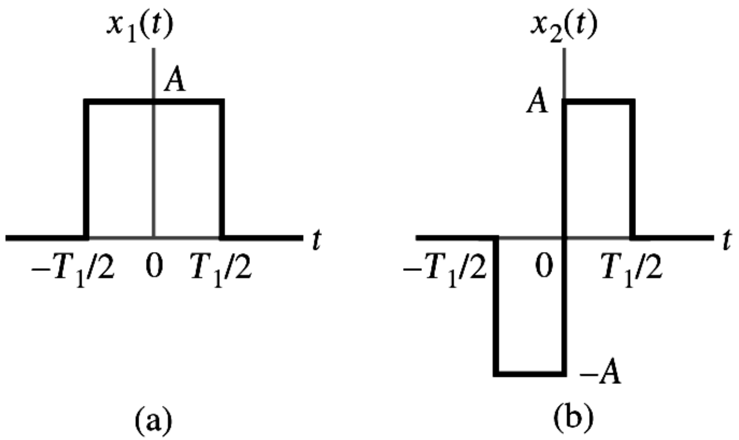

Figure 1.13 (p. 20)
(a) One example of continuous-time signal.
(b) Another example of a continuous-time signal.

1.4. Signal Classification

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◆ Example of periodic and nonperiodic signals: Fig. 1-14.

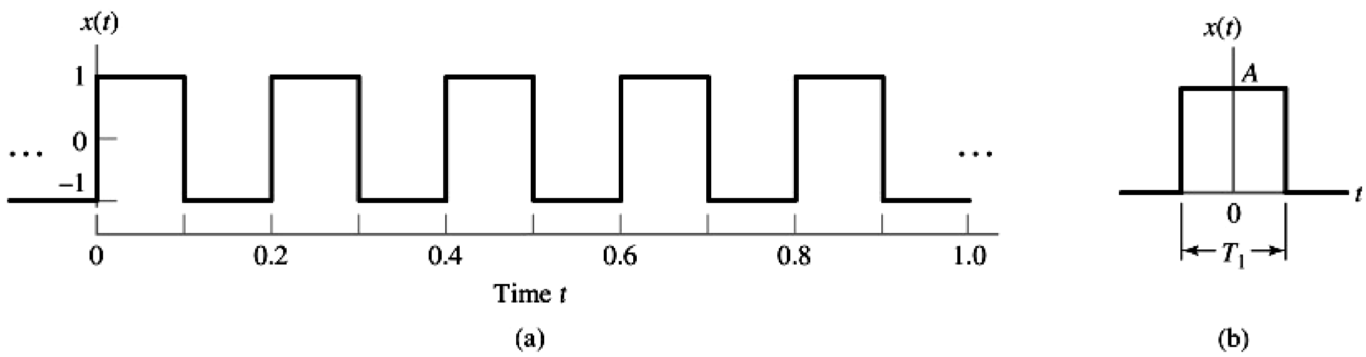


Figure 1.14 (p. 21)
(a) Square wave with amplitude A = 1 and period T = 0.2s.
(b) Rectangular pulse of amplitude A and duration T_1.

◆ Periodic and nonperiodic signals (Discrete-Time Case)

$$x[n] = x[n+N] \quad \text{for} \quad \text{integer} \quad n \qquad (1.10)$$

N = positive integer

Fundamental frequency of x[

$$\Omega = \frac{2\pi}{N} \qquad (1.11)$$

1.4. Signal Classification

$\frac{1}{T} = f$ cycles/s

Hertz

$T = .1$

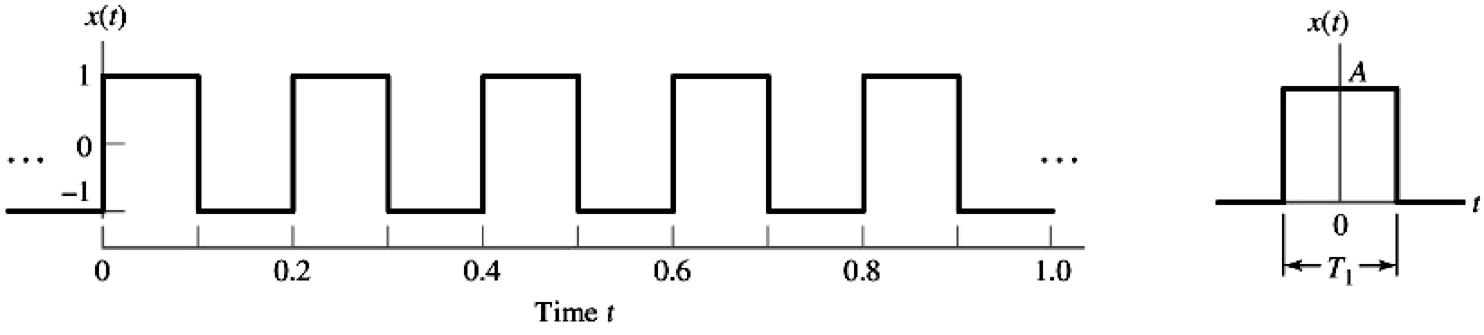
DT: $x(n) = x(n+N)$

$N =$ period smallest

$F = \frac{1}{N}$

1.4. Signal Classification

◆ **Example of periodic and nonperiodic signals: Fig. 1-14.**



(a) (b)

Figure 1.14 (p. 21)
(a) Square wave with amplitude A = 1 and period T = 0.2s.
(b) Rectangular pulse of amplitude A and duration T_1.

◆ **Periodic and nonperiodic signals (Discrete-Time Case)**

$$x[n] = x[n+N] \quad \text{for} \quad \text{integer} \quad n \qquad (1.10)$$

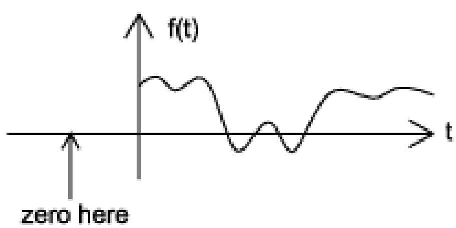
N = positive integer

Fundamental frequency of
***x*[**

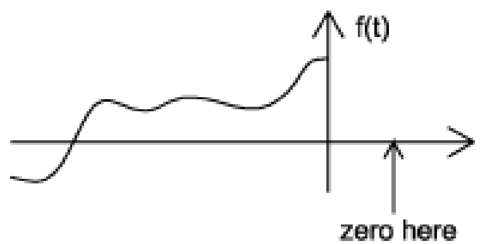
$$\Omega = \frac{2\pi}{N} \qquad (1.11)$$

1.4. Signal Useful Backgrounds

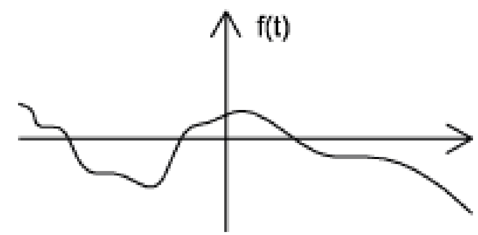
- **Causal and anticausal Signals**
  - Causal signals: zero for all negative time
  - Anti-causal signals: zero for all positive time
  - Non-causal : nonzero values in both positive and negative time



causal signal



anticausal signal

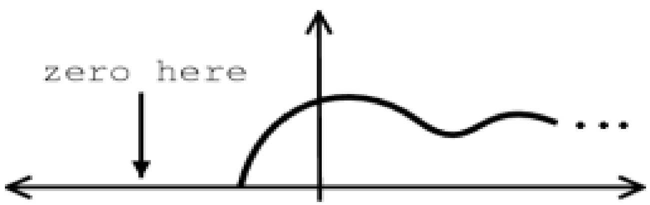


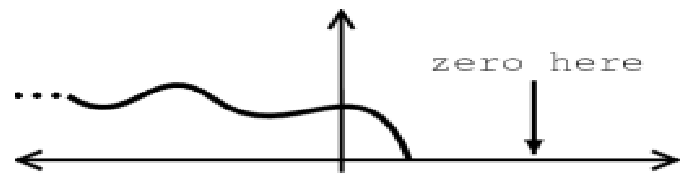
noncausal signal

1.4. Signal Useful Backgrounds

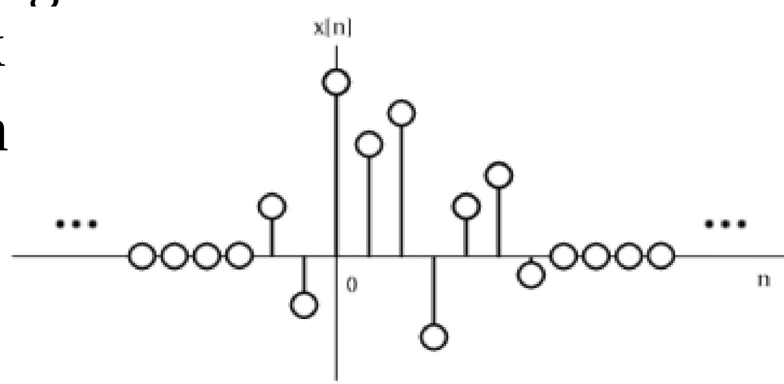
- Right-handed and left-handed Signals

  Right-handed and left handed-signal: zeros between a given variable and positive or negative infinity





- Finite and infinite length
  - Finite-length signal : nonzero over a finite interval tmin< t< tmax
  - Infinite-length umbers



1.4. Useful fomulae

- Euler's Formula

$$e^{j\theta} = \cos\theta + j\sin\theta$$

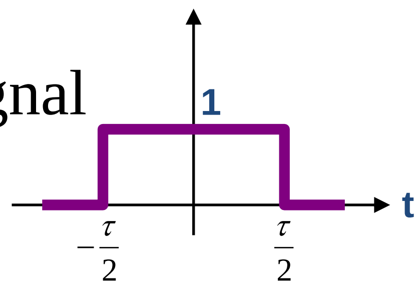
$$e^{-j\theta} = \cos\theta - j\sin\theta$$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

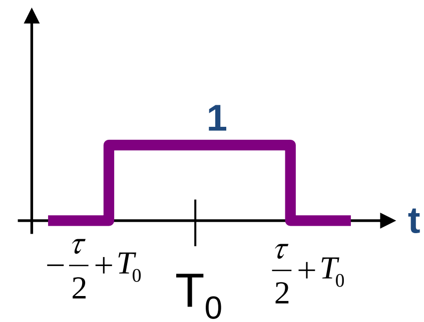
$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

- Rectangular Signal

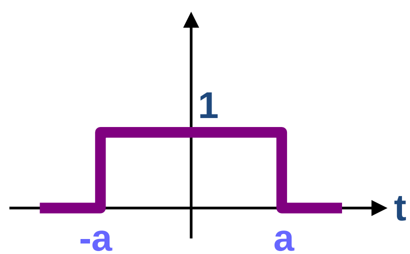
$$\mathrm{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| < \frac{\tau}{2} \\ 0, & |t| > \frac{\tau}{2} \end{cases}$$



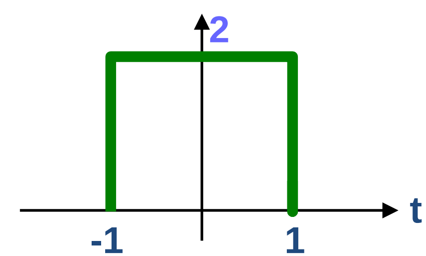
$$\mathrm{rect}\left(\frac{t - T_0}{\tau}\right)$$



$$\mathrm{rect}\left(\frac{t}{2a}\right)$$



$$2\mathrm{rect}\left(\frac{t}{2}\right)$$



1.4. Useful fomulae

1. $e^{j\theta} = \cos\theta + j\sin\theta$ (Euler's formula)

2. $\cos\theta = \frac{1}{2}\left(e^{j\theta} + e^{-j\theta}\right)$ (Inverse Euler formula)

3. $\sin\theta = \frac{1}{2j}\left(e^{j\theta} - e^{-j\theta}\right)$ (Another Inverse Euler formula)

4. $1 = e^{j2\pi} = e^{j2\pi n}$ for any integer n

5. $-1 = e^{j\pi} = e^{-j\pi}$

6. $(-1)^n = e^{j\pi n}$

7. $j = e^{j\pi/2}$

8. $-j = \frac{1}{j} = e^{-j\pi/2}$

1.4. Useful fomulae

$$\cos(\theta)\cdot\cos(\theta) = \left(\frac{e^{j\cdot\theta} + e^{-j\cdot\theta}}{2}\right)^2 = \frac{e^{j\cdot 2\theta} + e^{-j\cdot 2\theta}}{4} + \frac{1}{2} = \frac{1}{2}\cdot(1 + \cos(2\cdot\theta))$$

$$\sin(\theta)\cdot\sin(\theta) = \left(\frac{e^{j\cdot\theta} - e^{-j\cdot\theta}}{2\cdot j}\right)^2 = \frac{e^{j\cdot 2\cdot\theta} - 2\cdot e^{j\cdot\theta}\cdot e^{-j\cdot\theta} + e^{-j\cdot 2\cdot\theta}}{-4} = \frac{1}{2}\cdot(1 - \cos(2\cdot\theta))$$

$$\cos(\theta)\cdot\sin(\theta) = \frac{e^{j\cdot\theta} + e^{-j\cdot\theta}}{2}\cdot\frac{e^{j\cdot\theta} - e^{-j\cdot\theta}}{2\cdot j} = \frac{e^{j\cdot 2\cdot\theta} - 1 + 1 - e^{-j\cdot 2\cdot\theta}}{4\cdot j} = \frac{1}{2}\cdot\sin(2\theta)$$

1.4. Useful formulae

$$\cos(\theta \pm \phi) = \frac{e^{j(\theta \pm \phi)} + e^{-j(\theta \pm \phi)}}{2} = \frac{e^{j\theta} e^{\pm j\phi} + e^{-j\theta} e^{\mp j\phi}}{2}$$

$$= \frac{1}{2}\begin{bmatrix} (\cos\theta + j\sin\theta)(\cos\phi \pm j\sin\phi) + \\ (\cos\theta - j\sin\theta)(\cos\phi \mp j\sin\phi) \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} \cos\theta\cos\phi + j\sin\theta\cos\phi \pm j\sin\phi\cos\theta \mp \sin\theta\sin\phi + \\ \cos\theta\cos\phi - j\sin\theta\cos\phi \mp j\sin\phi\cos\theta \mp \sin\theta\sin\phi \end{bmatrix}$$

$$= \cos\theta\cos\phi \mp \sin\theta\sin\phi$$

1.5. Basic Operations on Signals

- Operations performed on dependent signals
- Operations performed on the independent signals

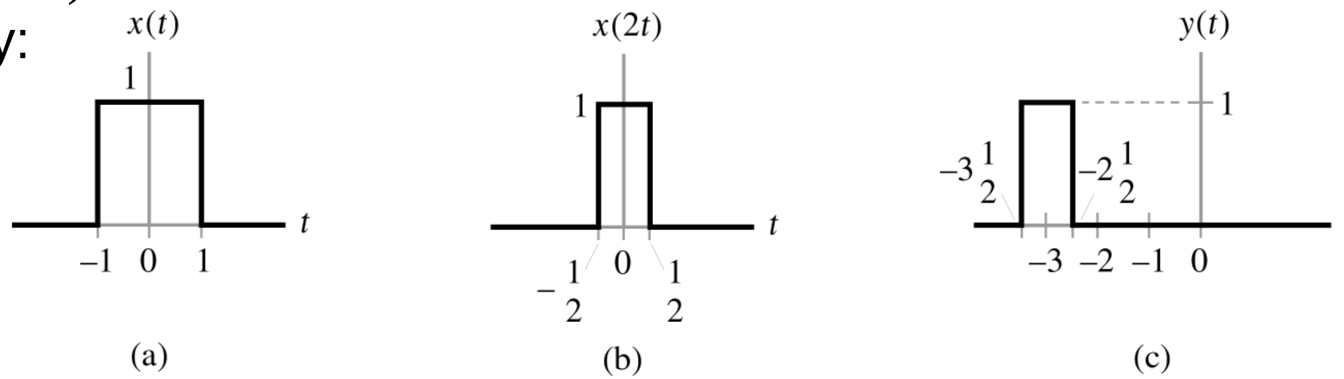
Operations performed on dependent signals

- Amplitude scaling $y(t) = cx(t)$
- Addition $y(t) = x_1(t) + x_2(t)$
- Multiplication $y(t) = x_1(t) \cdot x_2(t)$
- Differentiation $y(t) = \frac{d}{dx} x(t)$
- Integration $y(t) = \int_{-\infty}^{t} x(\tau) d\tau$

Operations performed on independent signals

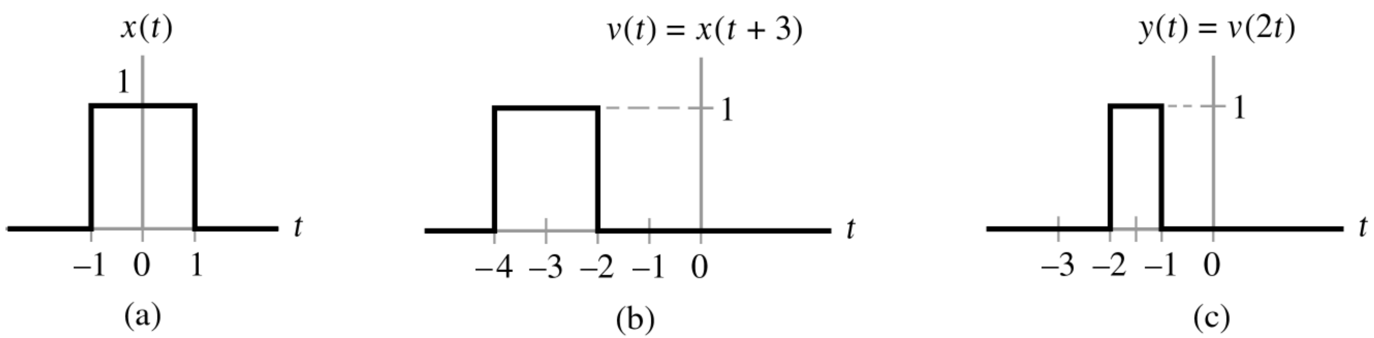
- $y(t) = x(2t + 3)$?

Incorrect way:



Time-scaled by 2 v(t) = x(2t). Shifted by 3 v(t) = x(2t) by 3

The proper way $y(t) = x(2t-3) = x\left(2(t-\frac{3}{2})\right)$



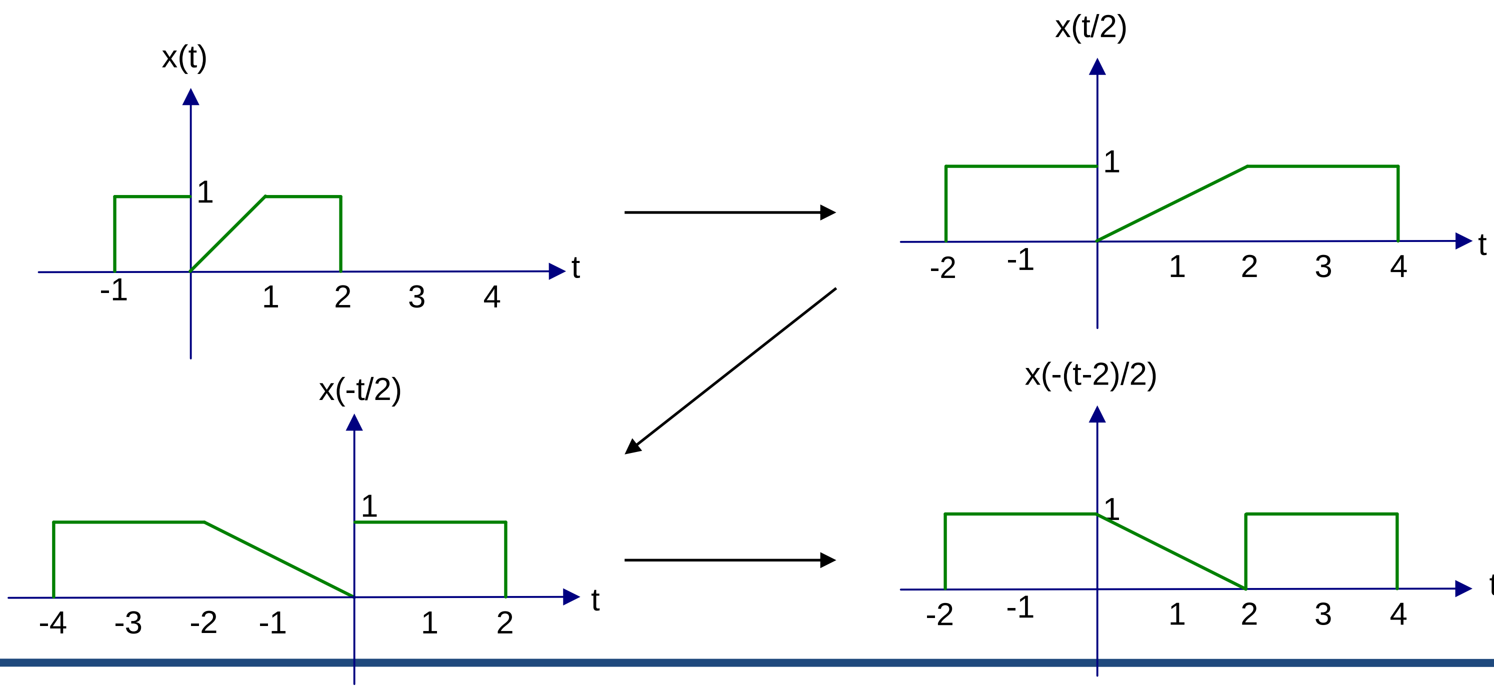
Shift by -3. Scaled by 2

Operations performed on independent signals

Ex) plot $x(t) \rightarrow x(1-\frac{t}{2})$

$$x(1-\frac{t}{2}) = x[-\frac{1}{2}(t-2)] = x[-\frac{(t-2)}{2}]$$

$$\begin{array}{ccccccc} & \text{scaling} & & \text{reversal} & & \text{shifting} & \\ & \downarrow & & \downarrow & & \downarrow & \\ x(t) & \rightarrow & x(\frac{t}{2}) & \rightarrow & x(-\frac{t}{2}) & \rightarrow & x[-\frac{(t-2)}{2}] \\ & \uparrow & & & \text{shr } 2 & & \\ & \text{expansion} & & & & & \end{array}$$



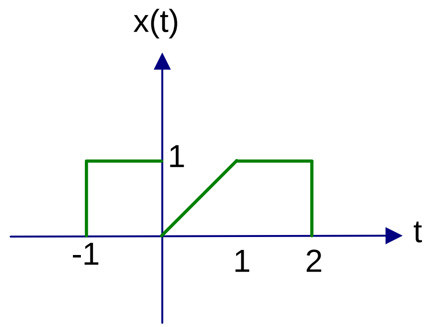
Operations performed on independent signals

- **Time & Amplitude Transformation**

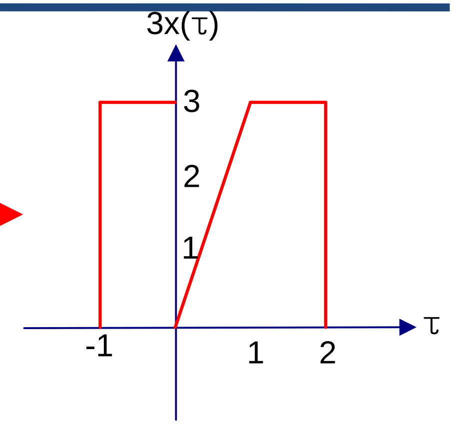
$$x_t(t) = 3x(1-\frac{t}{2})-1$$

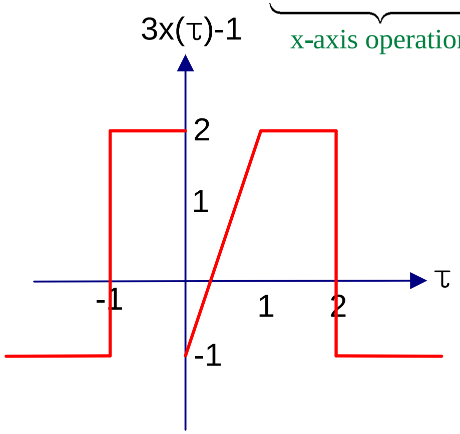
$$x(t) \Rightarrow x(\tau) \Rightarrow \underbrace{3x(\tau) \Rightarrow 3x(\tau)-1}_{\text{y-axis operation}}$$

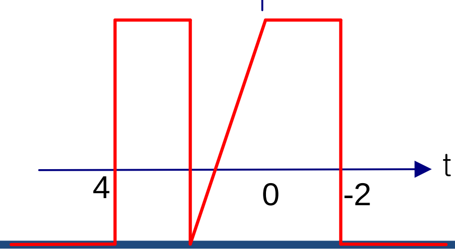
$$\Rightarrow \underbrace{3x(1-\frac{t}{2})-1}_{\text{x-axis operation}}$$



x(τ)

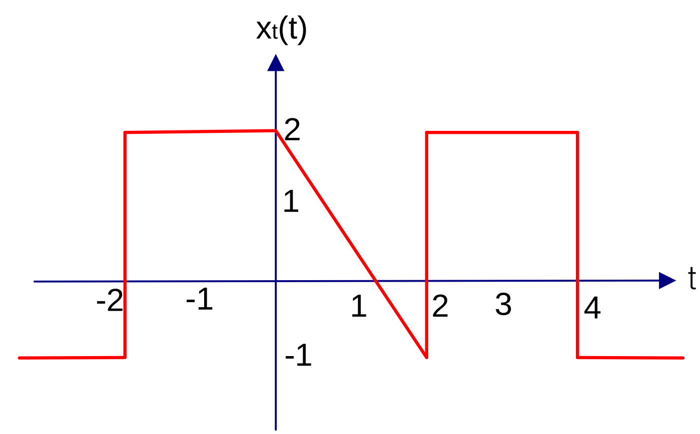






$$\tau = 1-\frac{t}{2}$$

$$t = 2-2\tau$$

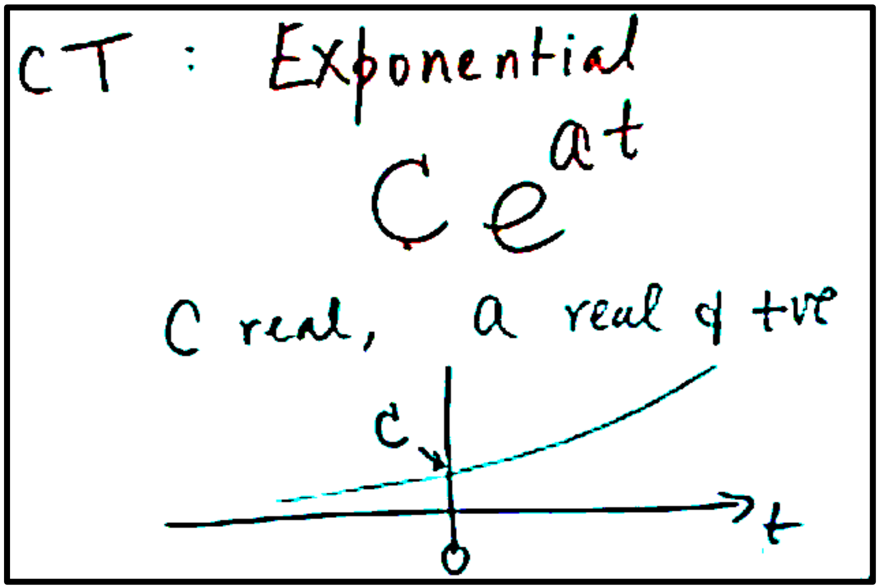


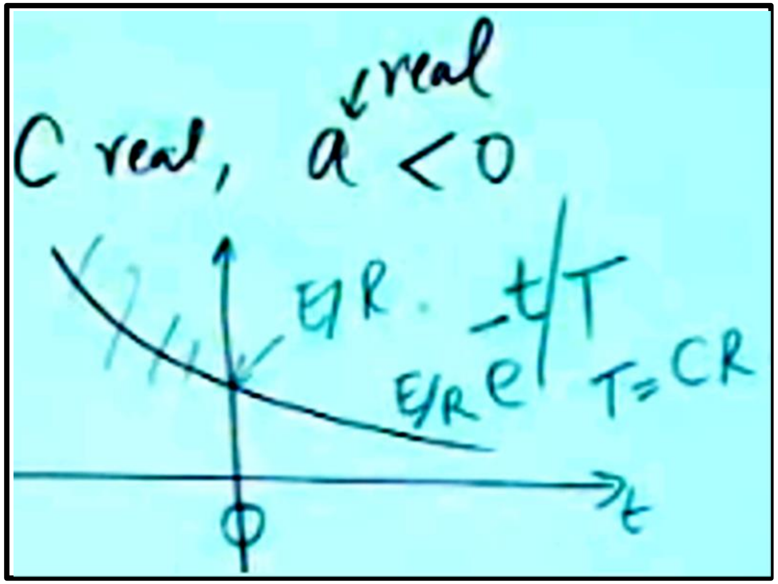
1.6. Elementary Signals

B and a are real parameters

★ 1.6.1 Exponential Signals $x(t) = Be^{at}$ (1.31)

1. **Decaying exponential, for which a < 0**
2. **Growing exponential, for which a > 0**



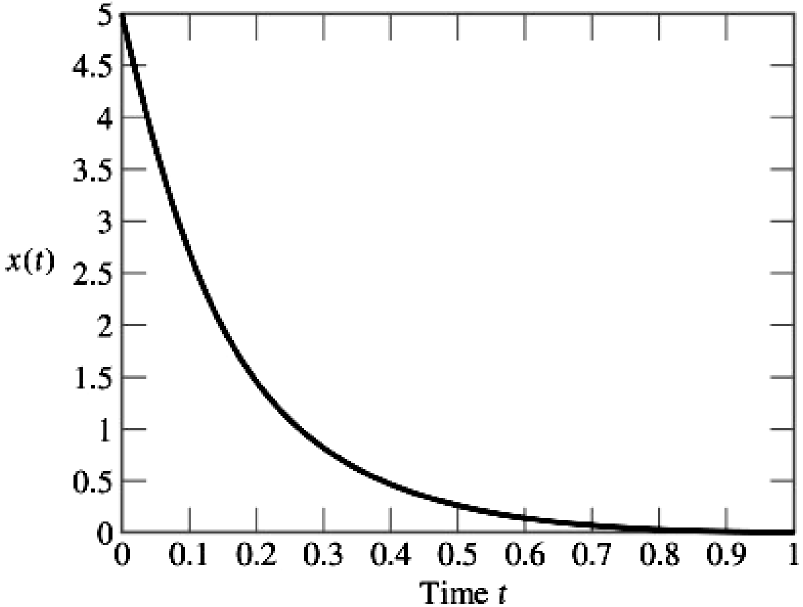


1.6. Elementary Signals

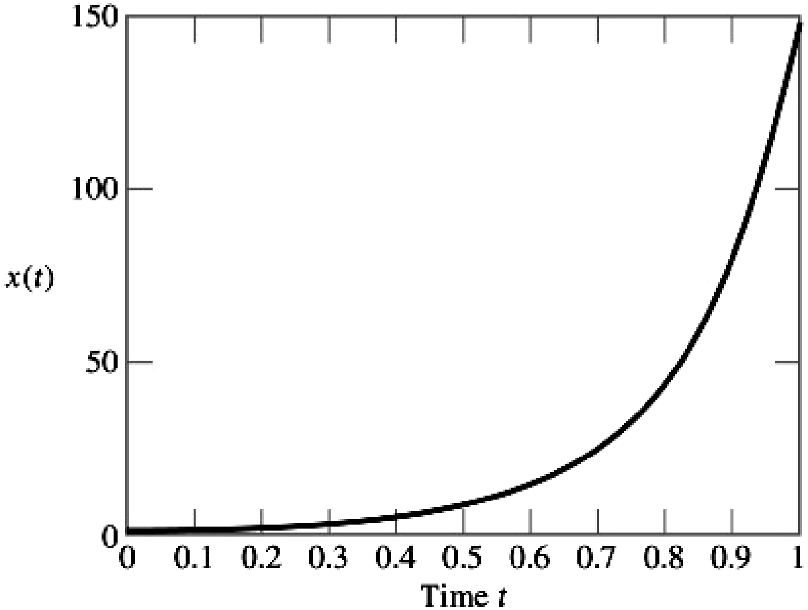
B and ***a*** are real parameters

★ 1.6.1 Exponential Signals $x(t) = Be^{at}$ (1.31)

1. **Decaying exponential, for which *a* < 0**
2. **Growing exponential, for which *a* > 0**



(a)



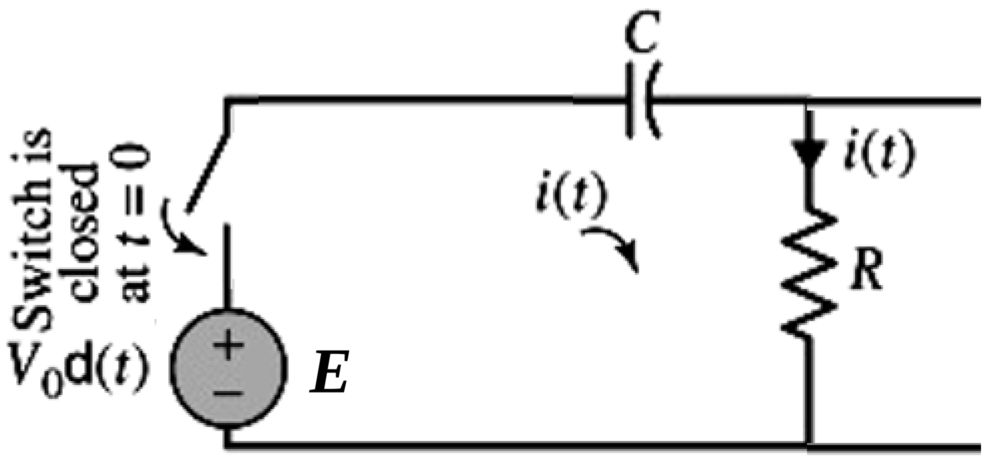
(b)

1.6. Elementary Signals

B and a are real parameters

★ 1.6.1 Exponential Signals $x(t) = Be^{at}$ (1.31)

1. **Decaying exponential, for which $a < 0$**
2. **Growing exponential, for which $a > 0$**



1.6. Elementary Signals

B and a are real parameters

★ 1.6.1 Exponential Signals $x(t) = Be^{at}$ (1.31)

1. **Decaying exponential, for which a < 0**
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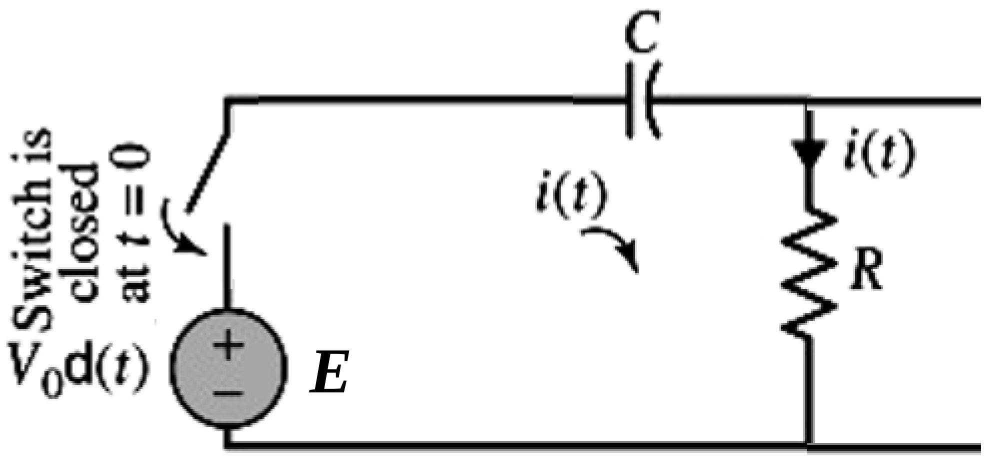
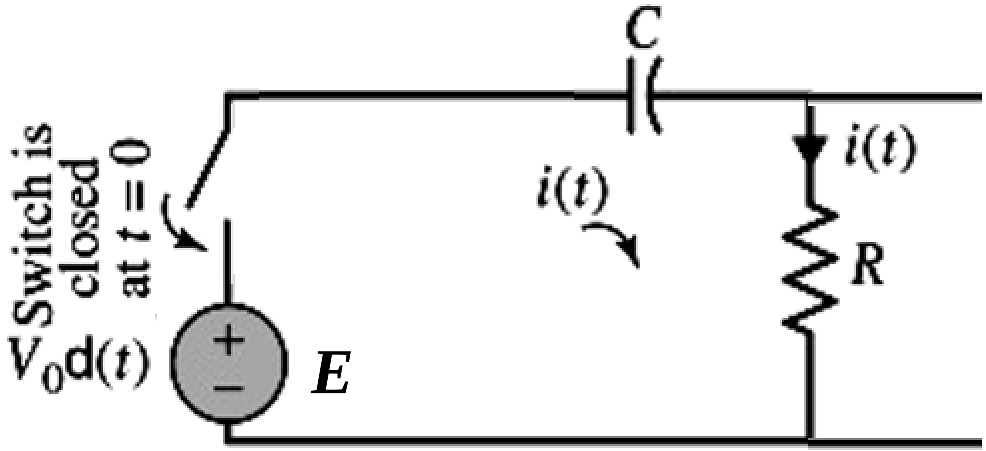


1.6. Elementary Signals



C real

$a = j\omega_0$

$e^{j\omega_0 t} \overset{?}{=} e^{j\omega_0 (t+T)}$

Periodic ?

$= e^{j\omega_0 t} e^{j\omega_0 T}$

$= 1$ if $\omega_0 T = 2\pi m$

$e^{j\omega_0 t}$ $\quad \omega_0 T = 2\pi$

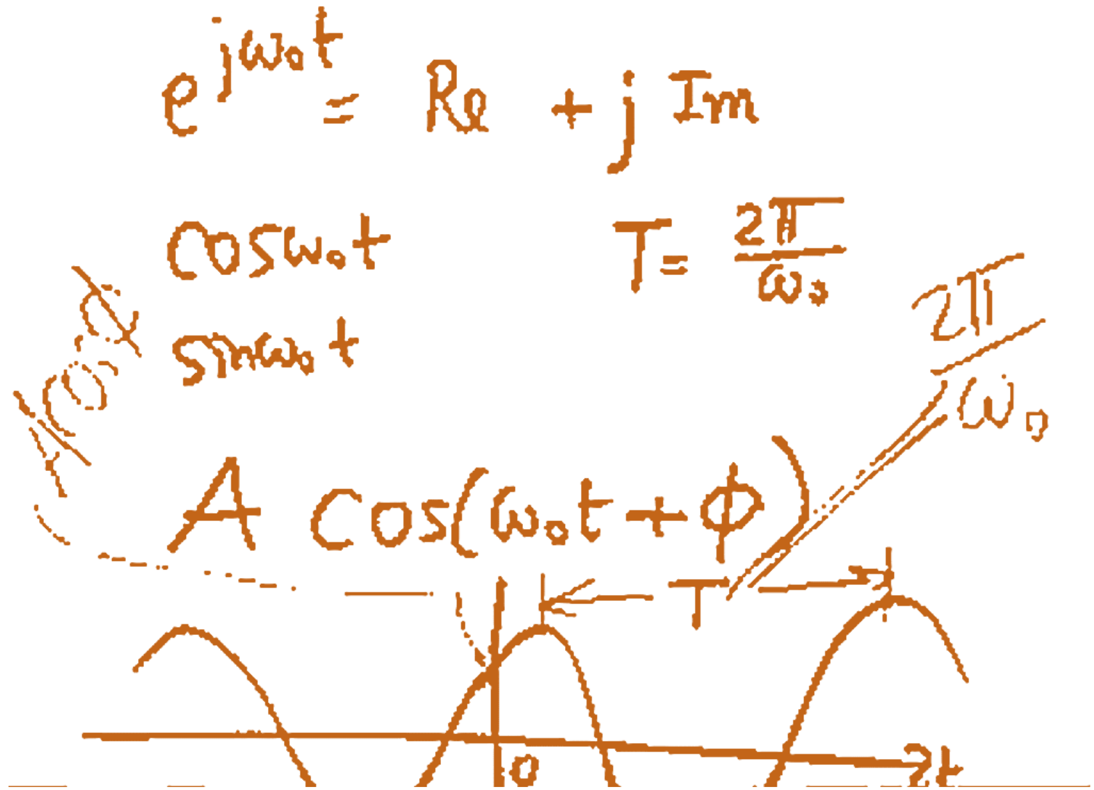
$T = \frac{2\pi}{\omega_0}$

$f = \frac{1}{T} = \frac{\omega_0}{2\pi}$

$\cos \omega_0 t = \mathrm{Re}\left[e^{j\omega_0 t}\right]$

$\sin \omega_0 t = \mathrm{Im}\left[e^{j\omega_0 t}\right]$

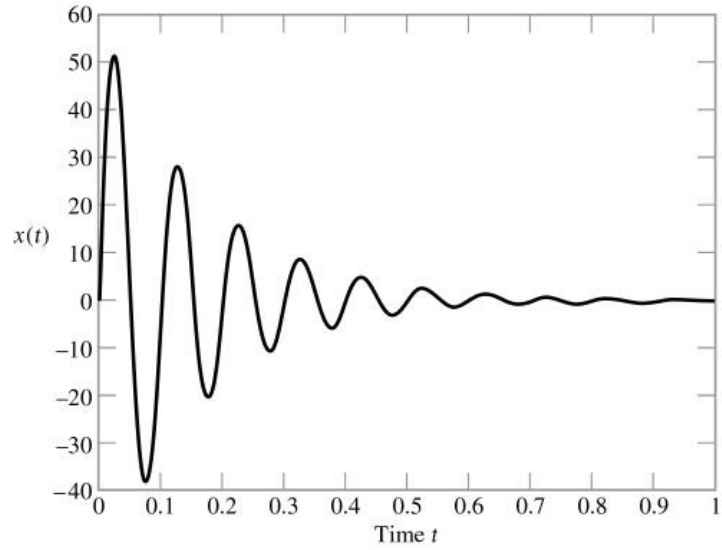
1.6. Elementary Signals



1.6. Elementary Signals

- **Exponential signals** $x(t) = Be^{at}$
- **Sinusoidal signals** $x(t) = A\cos(\varpi t + \phi)$
- **Exponentially damped sinusoidal signals**

$$x(t) = Ae^{at}\cos(\varpi t + \phi)$$



1.6. Elementary Signals

- **Exponential signals** $x(t) = Be^{at}$
- **Sinusoidal signals** $x(t) = A\cos(\varpi t + \phi)$
- **Exponentially damped sinusoidal signals**

$$x(t) = Ae^{at}\cos(\varpi t + \phi)$$

$x_1(t) = e^{j\omega_0 t}$

$x_2(t) = \underbrace{e^{jm\omega_0 t}}_{\text{m-th harmonic of } x_1}$ → integer

x_1 & x_2 are Harmonically related

$f_1 = \frac{\omega_0}{2\pi}$

$f_2 = \frac{m\omega_0}{2\pi} = m f_1$

CT: Exp. Signal

$c e^{at}$

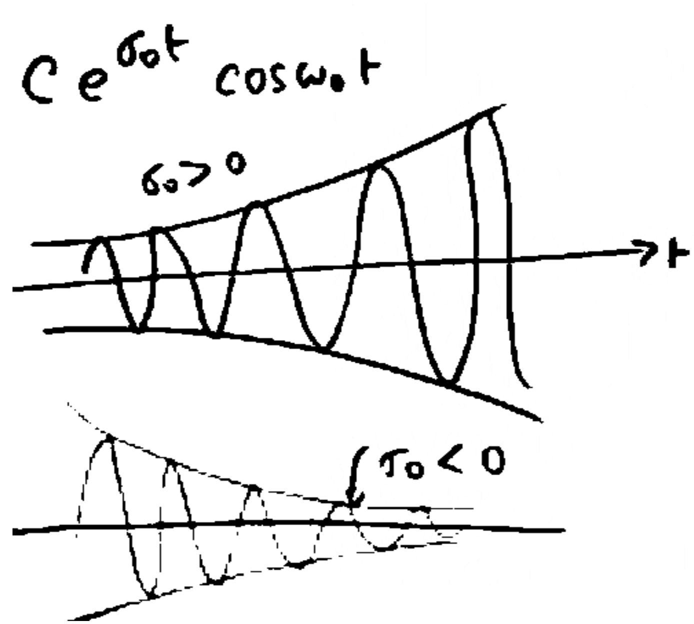
$a = (\sigma_0 + j\omega_0)t$

$Re[\, c e^{\sigma_0 t}\, \underline{e^{j\omega_0 t}}\,]$

$= c e^{\sigma_0 t} \cos\omega_0 t$

1.6. Elementary Signals

- **Exponential signals** $x(t) = Be^{at}$
- **Sinusoidal signals** $x(t) = A\cos(\varpi t + \phi)$
- **Exponentially damped sinusoidal signals**

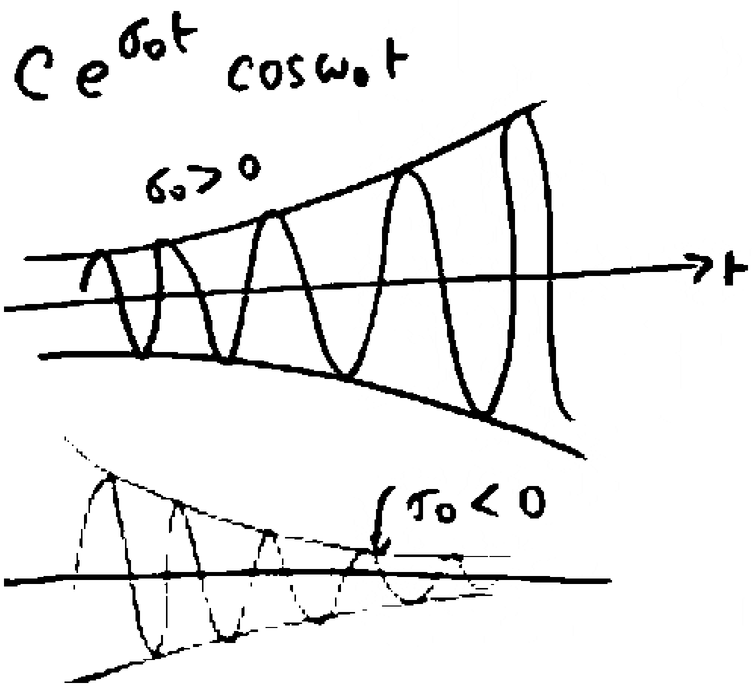


$$x(t) = Ae^{at}\cos(\varpi t + \phi)$$

1.6. Elementary Signals

- **Exponentially damped sinusoidal signals**

$$x(t) = Ae^{at}\cos(\varpi t + \phi)$$



$$C = C_1 + jC_2$$
$$= |C|e^{j\theta} \qquad \theta = \tan^{-1}\frac{C_2}{C_1}$$
$$a = \sigma_0 + j\omega_0$$
$$x(t) = Ce^{at} = \underline{|C|e^{\sigma_0 t}e^{j(\omega_0 t + \theta)}}$$

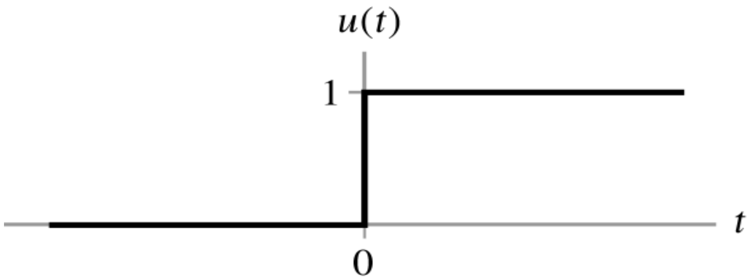
$\sigma_0 > 0$ — Growing

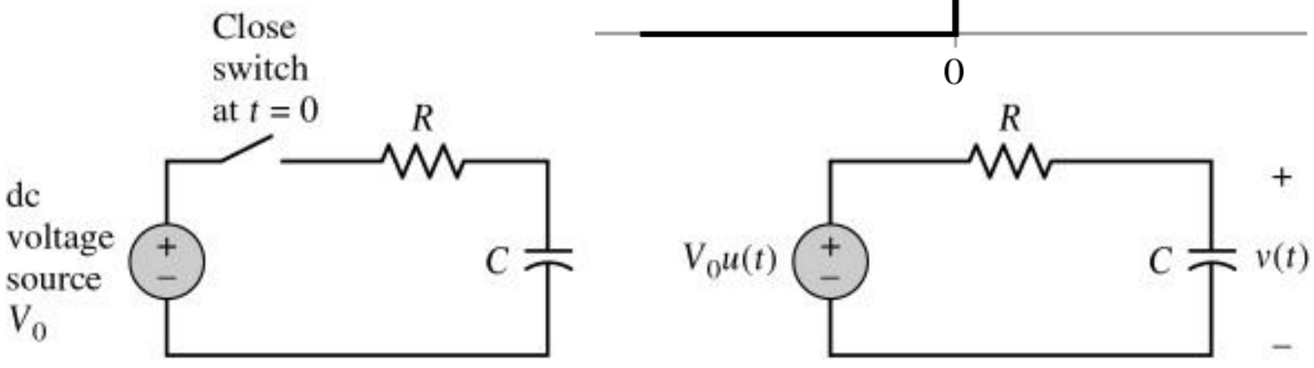
$\sigma_0 < 0 \longrightarrow$ Decaying

1.6. Elementary Signals

- **Step function**

$$x(t) = u(t)$$





(a) (b)

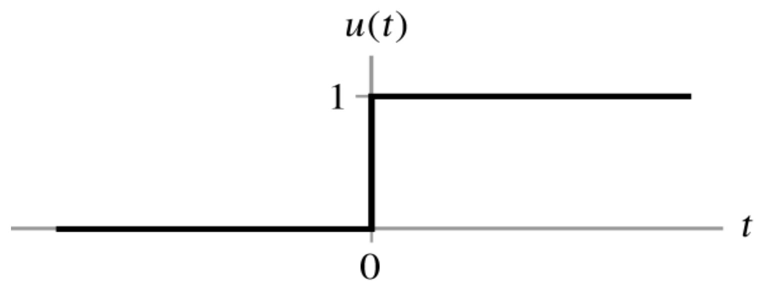
CT: Exponential

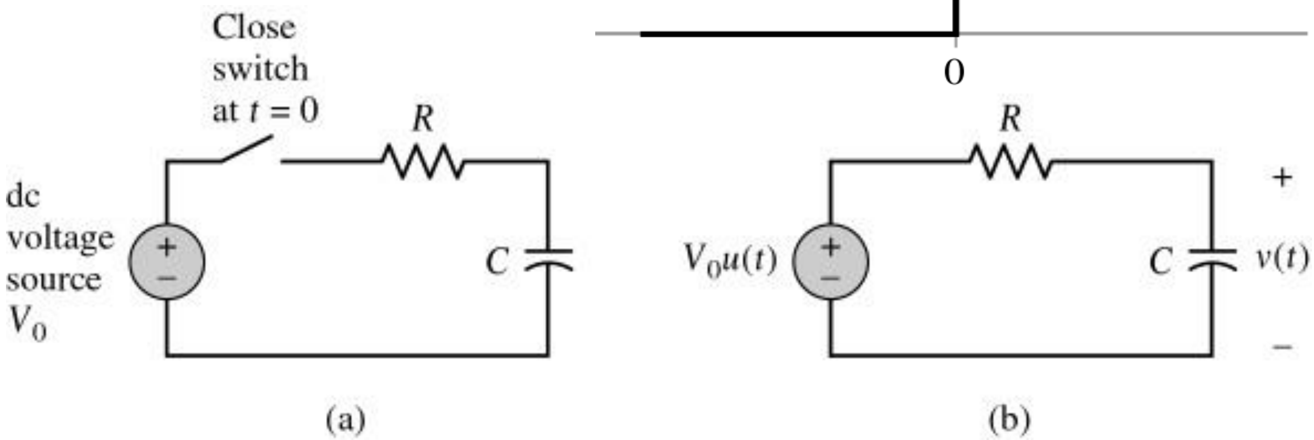
Unit step, $u(t) = \begin{cases} 0, & t<0 \\ 1, & t>0 \end{cases}$

1.6. Elementary Signals

- **Step function**

$$x(t) = u(t)$$



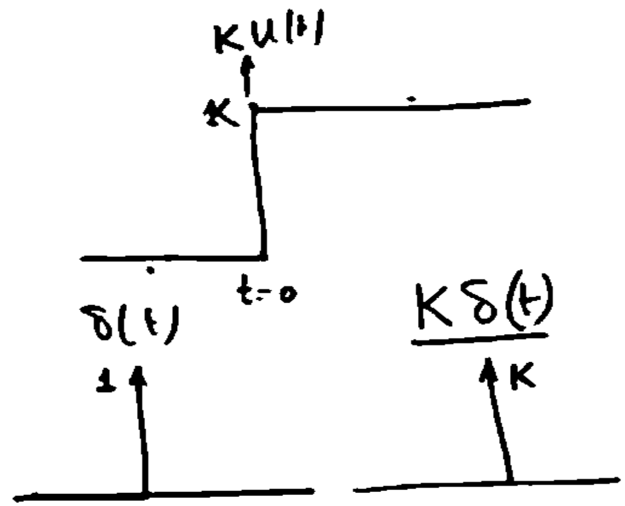


$$\int_{-\infty}^{t} \delta(\tau)\,d\tau = 0 \qquad t \le 0^-$$

$$= 1 \qquad t \ge 0^+$$

$$\int_{-\infty}^{t} \delta(\tau)\,d\tau = u(t) \qquad t > 0$$

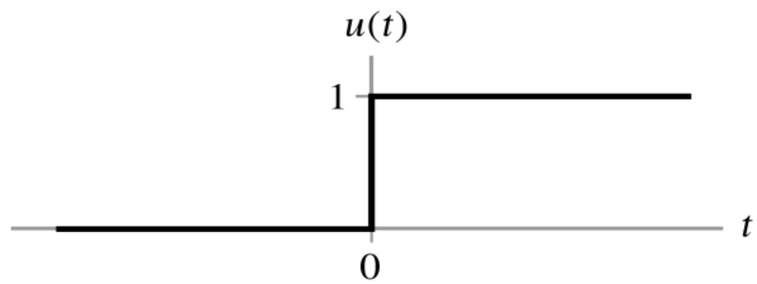
$$\frac{du(t)}{dt} = \delta(t)$$

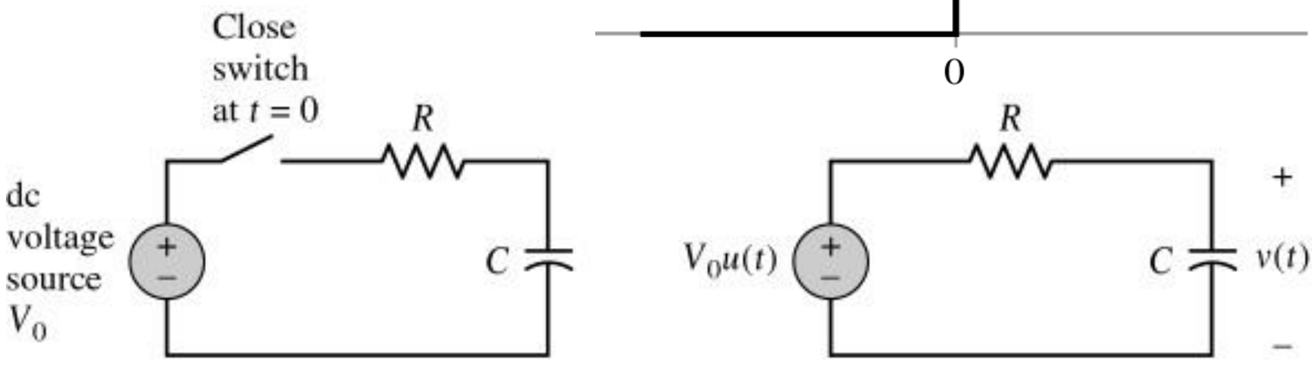


1.6. Elementary Signals

- **Step function**

$$x(t) = u(t)$$

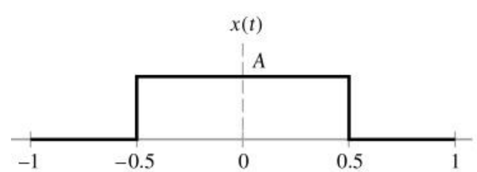




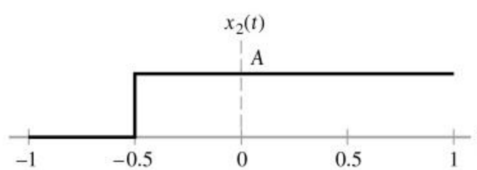
(a) (b)

- **Rectangular function**

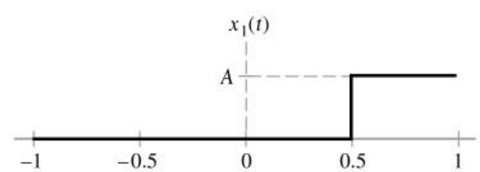
$$x(t) = rect(t)$$



(a)



(c)



(b)

$$x(t) = u(t + 1/2)$$

$$x(t) = u(t - 1/2)$$

1.6. Elementary Signals

Impulse function

: $\int_{-\infty}^{\infty} \delta(t)dt = 1$ or, Dirac delta

$\delta(t) = 0, t \neq 0$

$\delta(t)$ is unbounded at t = 0

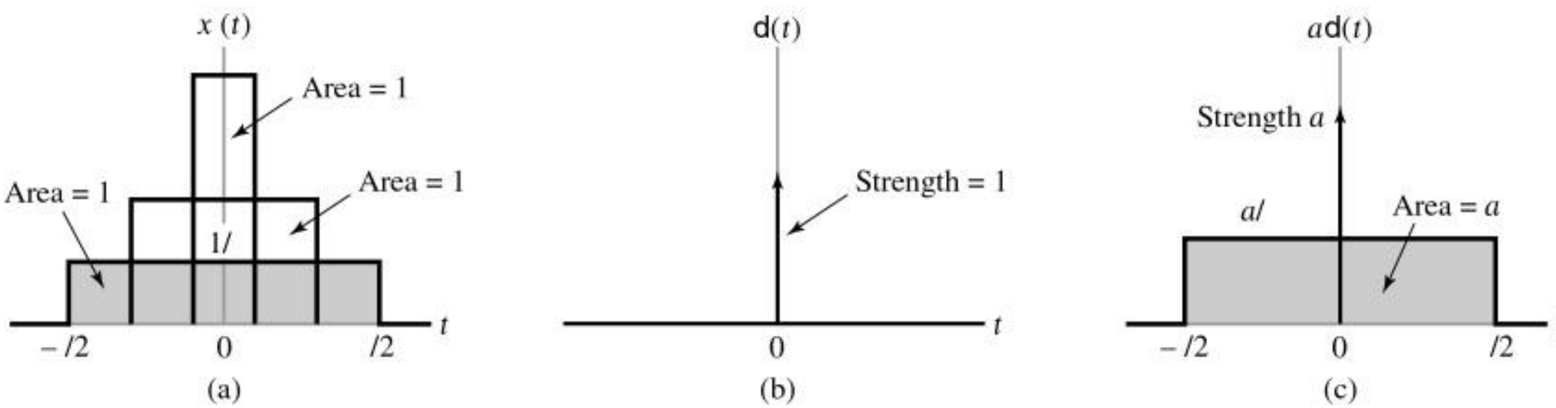
$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt = x(t_0)$ → time sifting

$\delta(t) = \delta(-t) \leftarrow$ even function

For discrete signal, Kronecker delta

$x[n] = \delta[n] = \delta_n$

$\delta[0] = 1$



1.6. Elementary Signals

- Properties of impulse function
  - Scaling
    $$\int \delta(at+b)dt = \frac{1}{|a|}\int \delta(t+\frac{b}{a})dt$$
  - Sampling
    $$\int x(t)\delta(t)dt \rightarrow \int x(0)\delta(t)dt$$
  - Sifting (no shifting)
    $$x(t)\delta(t-t_o) = x(t_o)\delta(t-t_o)$$
    $$\int x(t)\delta(t-t_o)dt = x(t_o)$$
    $$\delta(t) = \begin{cases} 0, & t \neq t_o \\ undefined, & t = t_o \end{cases}$$

1.6. Elementary Signals

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ex)

ⓐ $\int_{-2}^{1}(t+t^2)\cdot\delta(t-3)dt=0$

ⓑ $\int_{-2}^{4}(t+t^2)\cdot\delta(t-3)dt=12$

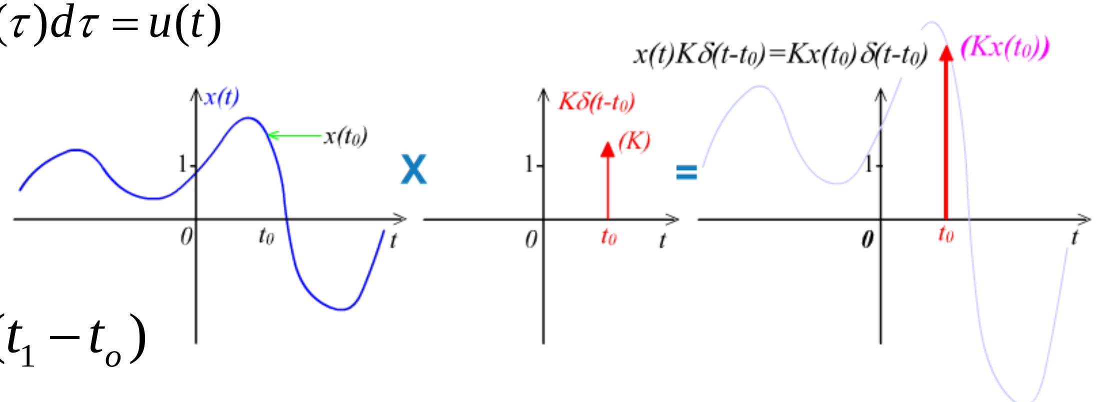
ⓒ $\int_{0}^{3}e^{t-2}\cdot\delta(2t-4)dt=\frac{1}{2}$

ⓓ $\int_{-\infty}^{t}\delta(\tau)d\tau=?$ $\quad \begin{Bmatrix} If & t<0, & 0 \\ If & t>0, & 1 \end{Bmatrix} \equiv u(t)$

Relationship between $\delta(t)$ & $u(t)$ (Integral $\leftrightarrow$ Derivative Pair)

ex) $\frac{du(t)}{dt}=\delta(t)$ $\qquad \int_{-\infty}^{t}\delta(\tau)d\tau=u(t)$

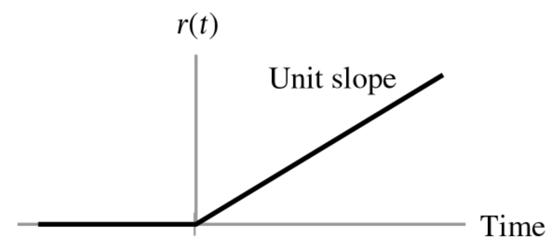
$Kx(t)\delta(t)?$ $\longrightarrow$



$$\int f(t-t_o)\delta(t-t_1)dt = f(t_1-t_o)$$

1.6. Elementary Signals

Ramp function

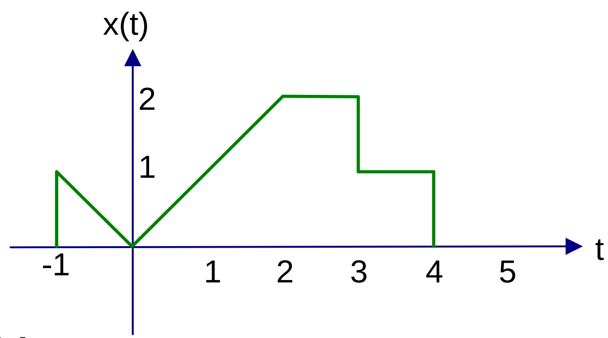


$$r(t)=\begin{cases} t, t \geq 0 \\ 0, t<0 \end{cases}$$

$$r(t)=\int_{-\infty}^{t} u(\tau)d\tau=\int_{0}^{t} d\tau \quad \leftarrow \text{Integration of Unit-Step Function}$$

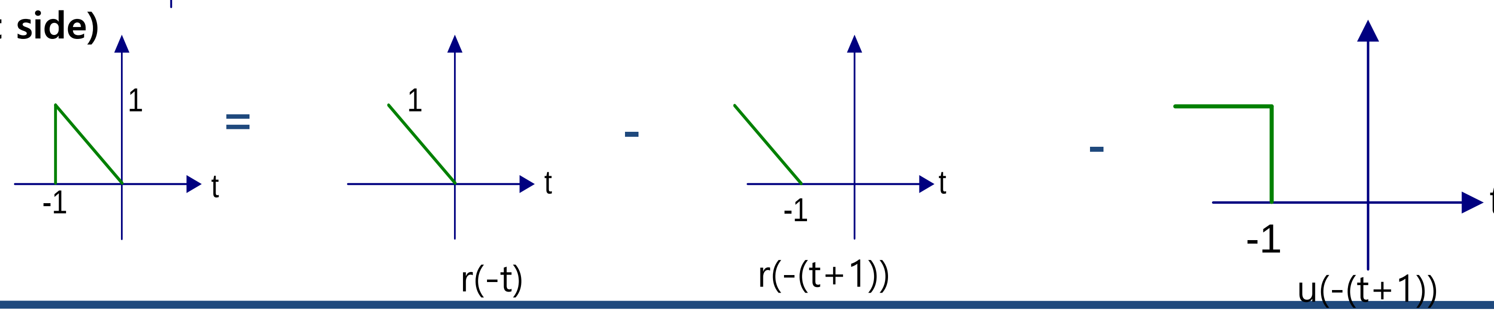
$$\int_{0}^{\alpha t} d\tau=\begin{cases} \alpha t, t \geq 0 \\ 0\ , t<0 \end{cases} \quad \leftarrow \text{slope of Ramp function} = \alpha$$

Ex)



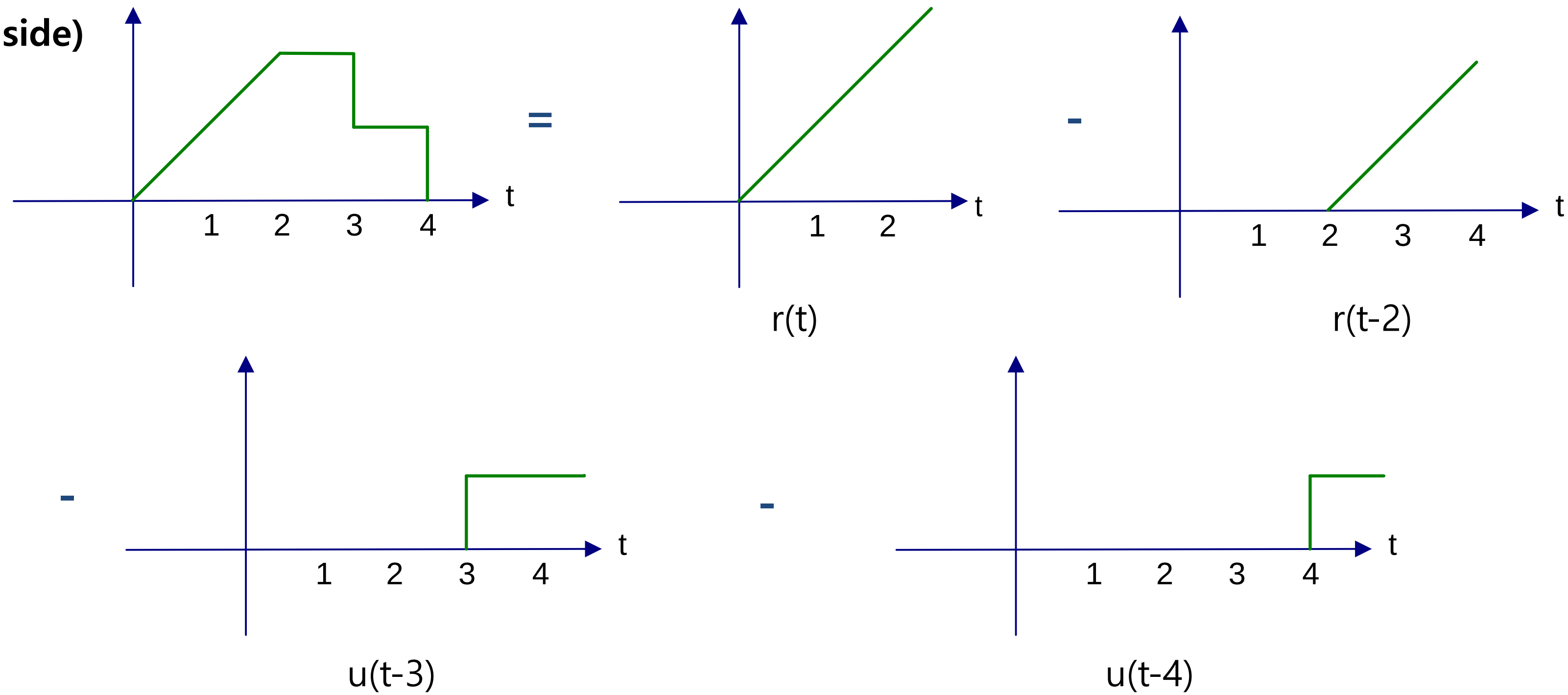
Deal left and right separately

(Left side)



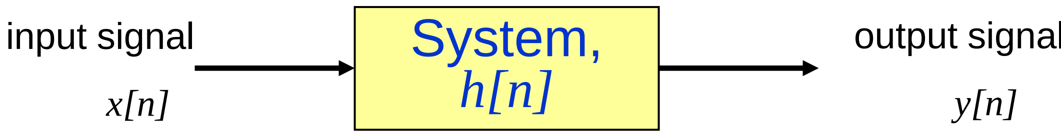
1.6. Elementary Signals

(Right side)

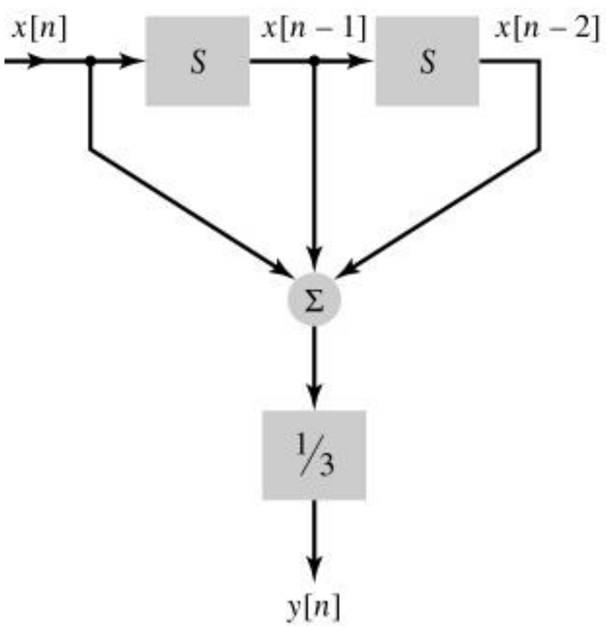


$$x(t) = r(-t) - r(-t-1) - u(-t-1) + r(t) - r(t-2) - u(t-3) - u(t-4)$$

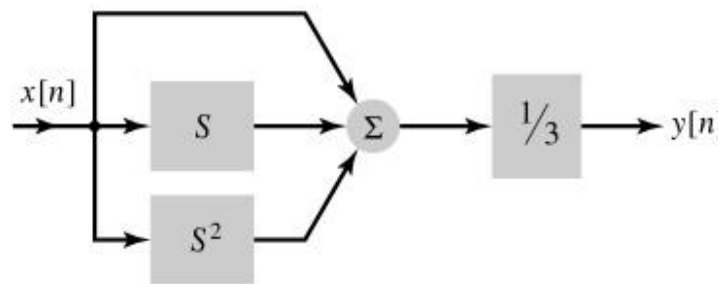
1.7. Systems Viewed as Interconnection of Operations



S : delay function



(a)



(b)

$$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$$

1.8. Properties of Systems

- Stability
- Memory
- Invertibility
- Time Invariance
- Linearity

Stability

- BIBO stable

A system is said to be *bounded-input bounded-output stable* if every bounded input results in a bounded output.

$$|x(t)| \le M_x < \infty, \quad \forall t, \rightarrow |y(t)| \le M_y < \infty, \quad \forall t$$

Ex) $\bullet\ |x(t)| \le M_x < \infty, \quad y[n] = \frac{1}{3}\left|x[n] + x[n-1] + x[n-2]\right|?$

$$y[n] \le \frac{1}{3}(|x[n]| + |x[n-1]| + |x[n-2]|)$$

$$\le \frac{1}{3}(M_x + M_x + M_x) = M_x, \quad \text{stable}$$

$\bullet\ r > 1, \quad y[n] = r^n x[n]?$

$$\left|y[n]\right| = \left|r^n x[n]\right|$$

$$= \left|r^n\right| \left|x[n]\right|$$

$$\le \left|r^n\right| M_x, \quad \text{unstable}$$

Stability

Ex) The collapse of the Tacoma Narrows suspension bridge on November 7, 1940.

(a) the twisting motion of the bridge's center span just before failure.

(b) A few minutes after the first piece of concrete fell, this second photograph shows a 600-ft section of the bridge breaking out of the suspension span and turning upside down as it crashed in Puget Sound, Washington. Note the car in the top right-hand corner of the photograph.

(a)

(b)

Memory

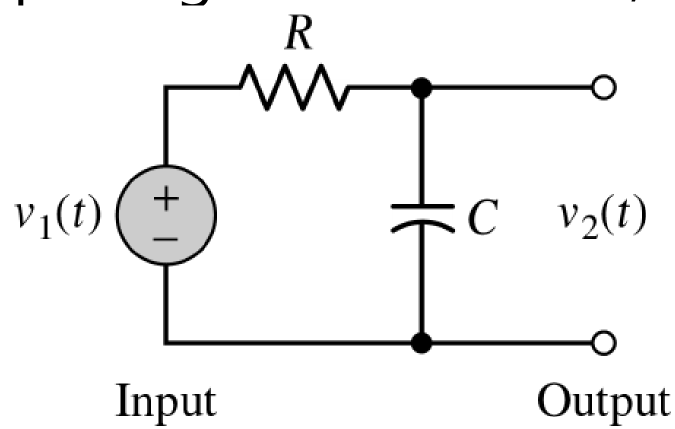
- Memory system

A system is said to possess *memory* if its output signal depends on *past values* of the input signal. Otherwise, Memoryless system

$$memoryless: -i(t) = \frac{1}{R} v(t)$$

ex) $$memory: -i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,d\tau$$

$$memory: -y[n] = x[n] + x[n-1]$$



$$memoryless: y(t) = x(t) + 4,$$

$$y(t) = (t+5)x(t)$$

$$memory: \quad y(t) = x(t+4)$$

$$y(t) = [x(t+7)]^2$$

$$y(t) = x(3)$$

$$y(t) = x(3t)$$

Causality

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- Causal system

A system is said to be *causal* if the present value of the output signal depends only on the *present* and/or *past* values of the input signal.

Ex)

$$y[n] = x[n] + x[n-1] \Rightarrow causal$$

$$y[n] = x[n+1] + x[n-1] \Rightarrow non-causal$$

$$y(t_o) = \int_{-\infty}^{t_O + a} x(t)dt \Rightarrow non-causal$$

$$y(t) = x(-t) \Rightarrow non-causal$$

memoryless system ➔ causal system,

but the causal system does NOT imply the memoryless system

Invertiblity

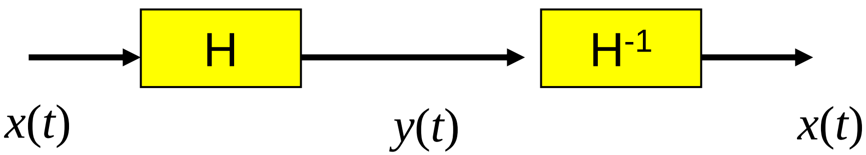
- Invertible system

A system is said to be *invertible* if the input of the system can be recovered from the system output.

H^{-1} is said to be an invertible system of H.

H: $x \rightarrow y$, H^{-1}: $y \rightarrow x$

$H^{-1}\{y(t)\} = H^{-1}\{H\{x(t)\}\}$, $H^{-1}H = I$



$$y(t) = \frac{1}{L}\int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow x(t) = L\frac{d}{dt}y(t)$$

$$y(t) = x^2(t)?$$

Time Invariance

- Time invariant system

A system is said to be *time-invariant* if a time delay or time advance of the input signal leads to an identical time shift in the output signal. $y_i(t) = H\{x(t-t_0)\}$

$$= H\{S^{t0}\{x(t)\}\} = HS^{t0}\{x(t)\}$$

$$y_0(t) = S^{t0}\{y(t)\}$$

$$= S^{t0}\{H\{x(t)\}\} = S^{t0}H\{x(t)\}$$

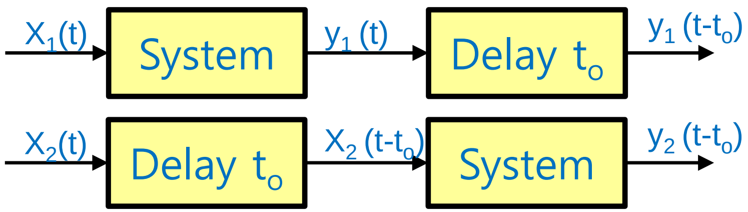
$\therefore HS^{t0} = S^{t0}H$ is required.

T.I.V. checking procedure:

-Let $x_1(t) \rightarrow y_1(t)$, then $y_1(t-t_0)$.

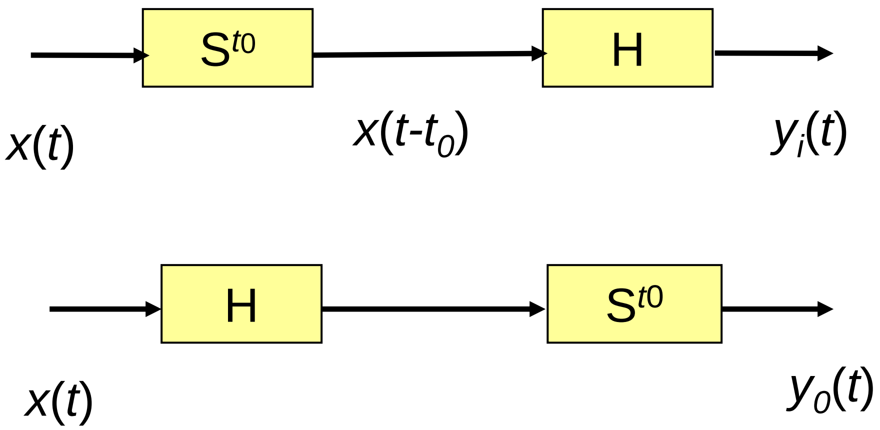
-Find $y_2(t)$ w.r.t. $x_2(t) = x_1(t-t_0)$,

and compare with $y_1(t-t_0)$



Time Invariance

- Are following two systems equivalent?



Time Invariance

ex)

$$y(t)=\int_0^t x(\tau)d\tau \quad \text{T.I.?}$$

$$\Rightarrow \text{ for } x_1(t), \qquad y_1(t)=X(\tau)\Big|_0^t = X(t)-X(0) \rightarrow y_1(t-t_o)=X(t-t_o)-X(0)$$

$$\text{for } x_1(t-t_o), \ \ y_2(t)=X(\tau-t_o)\Big|_0^t = X(t-t_o)-X(-t_o) \neq y_1(t-t_o), \quad \text{T.V}$$

ex)

$$y(t)=\int_0^T x(t-\tau)d\tau \quad \text{T.I.?}$$

$$\Rightarrow \text{ for } x_1(t), \qquad y_1(t)=X(t-\tau)\Big|_0^T = X(t-T)-X(t) \rightarrow y_1(t-t_o)=X(t-t_o-T)-X(t-t_o)$$

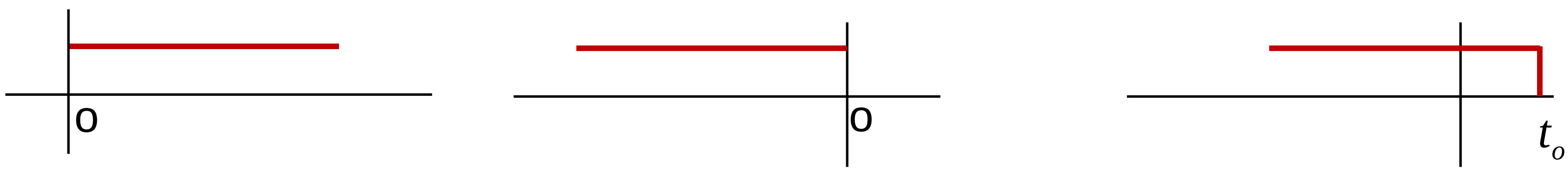
$$\text{for } x_1(t-t_o), \ \ y_2(t)=X(t-t_o-\tau)\Big|_0^T$$

$$= X(t-t_o-T)-X(t-t_o) = y_1(t-t_o), \quad \text{T.I}$$

Time Invariance

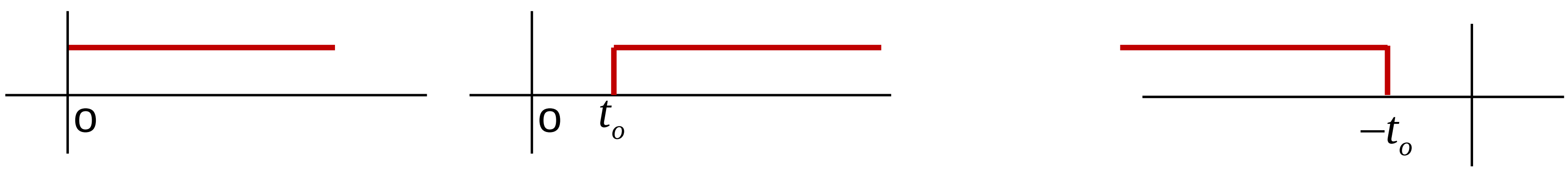
ex) $y(t)=x(-t)$ T.I.?

$x(t)=u(t)$ → **Time reversal** → $y(t)=x(-t)=u(-t)$ → **Delay t_o** → $y(t-t_o)=x(-(t-t_o))=u(t_o-t)$



$\neq$

$x(t)=u(t)$ → **Delay t_o** → $x(t-t_o)=u(t-t_o)$ → **Time reversal** → $y_d(t)=x(-t-t_o)=u(-t-t_o)$



Linearity

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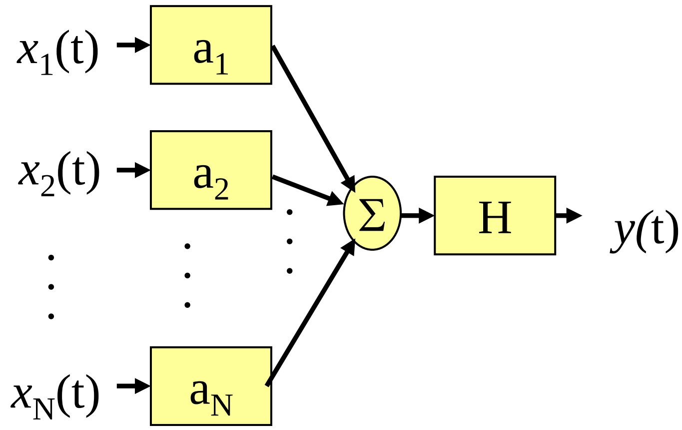
- Linear system :

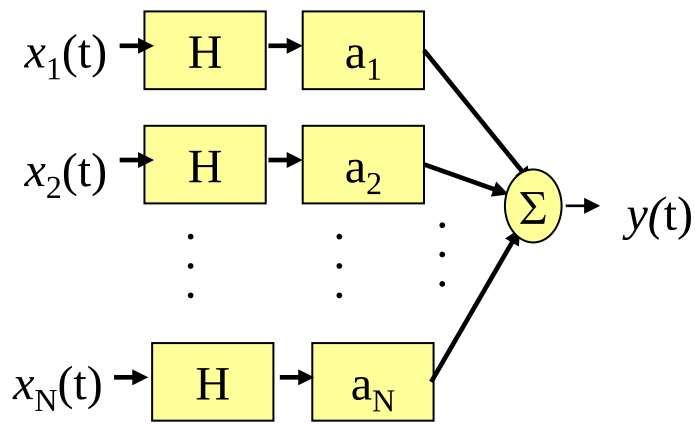
A system is said to be *linear* if it satisfies the *principle of superposition*.

$$\text{let } x(t) = \sum_{i=1}^{N} a_i x_i(t)$$

$$\begin{aligned} y(t) &= H\{x(t)\} \\ &= H\left\{\sum_{i=1}^{N} a_i x_i(t)\right\} \\ &= \sum_{i=1}^{N} a_i H\{x_i(t)\} \\ &= \sum_{i=1}^{N} a_i y_i(t), \end{aligned}$$

where $y_i(t) = H\{x_i(t)\}$





Linearity

Ex) • $y[n] = nx[n]$

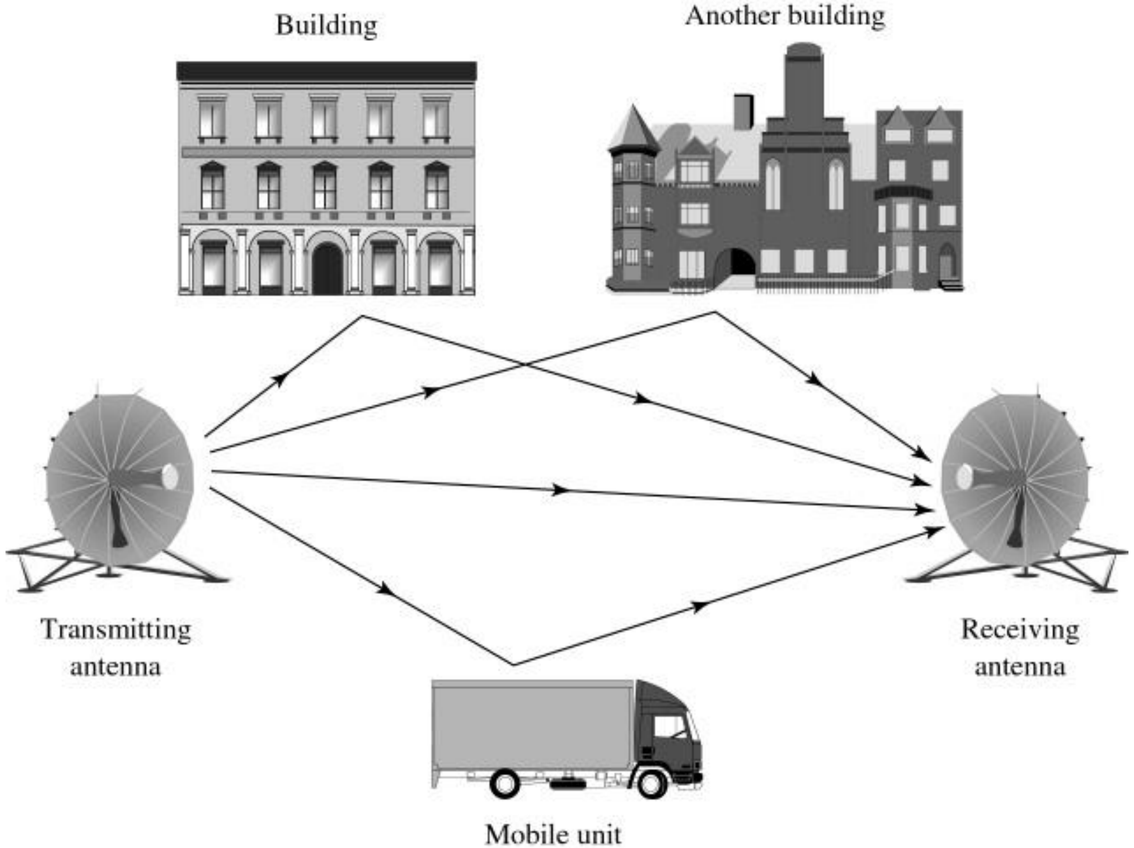
• $y(t) = x(t)x(t-1)$

Check superposition with simple two inputs.

$$x(t) = a_1 x_1(t) + a_2 x_2(t)$$

Theme Examples

Physical phenomenon:
Example of multiple propagation paths in a wireless communication environments.



Theme Examples

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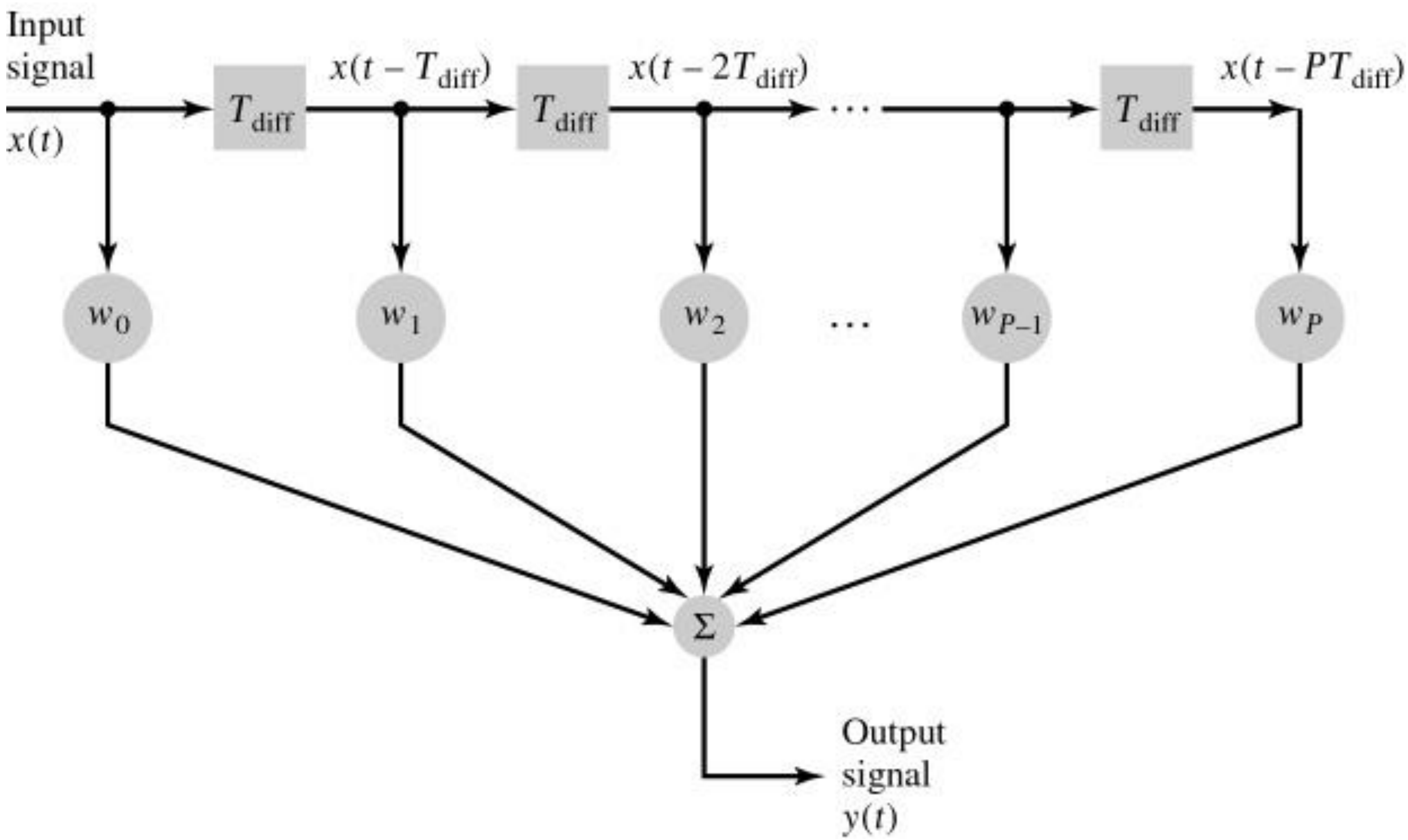
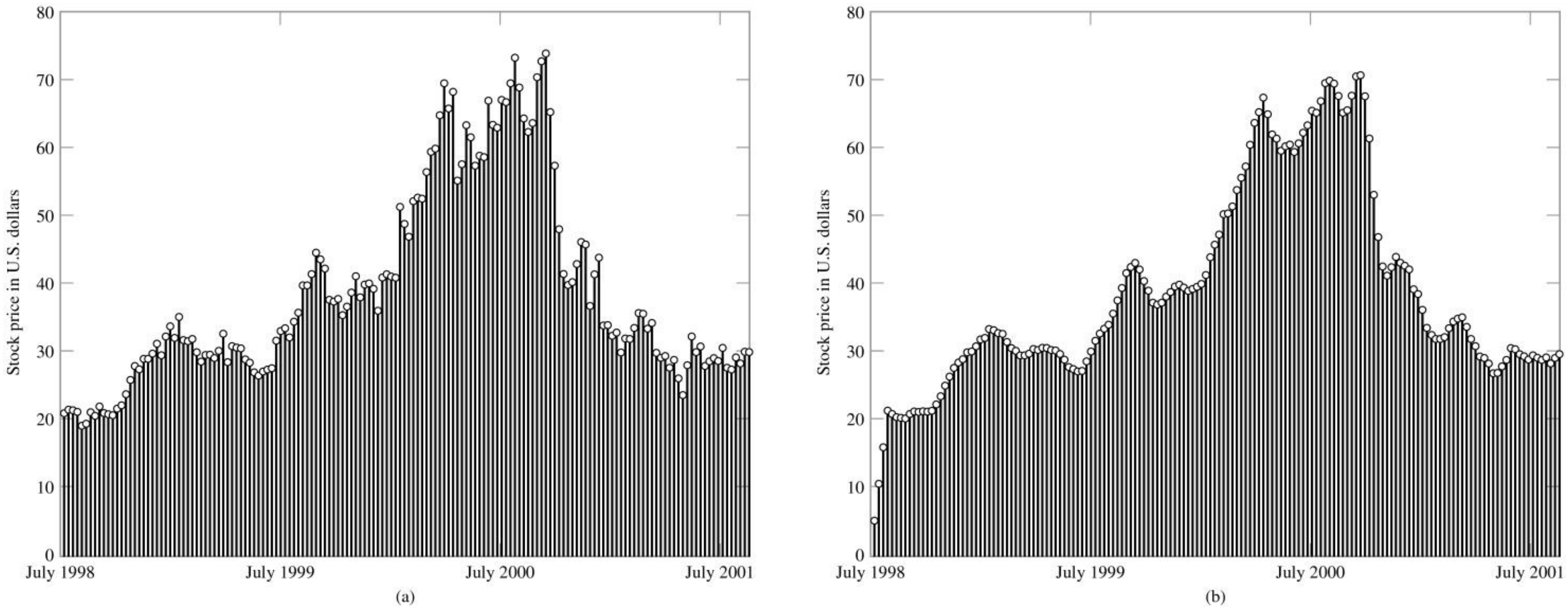
Mathematical modeling

Tapped-delay-line model of a linear communication channel, assumed to be time-invariant



Stock Price : filtering

(a) Fluctuations in the closing stock price of Intel over a three-year period.
(b) Output of a four-point moving-average system.



Brief break (if on schedule)

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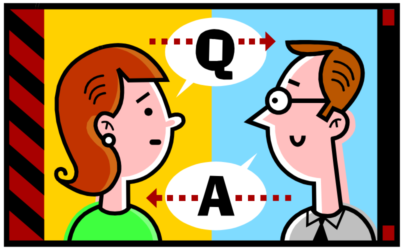
Q&A?



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